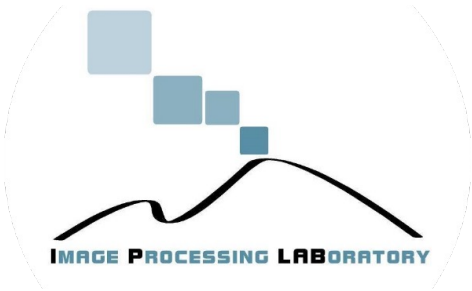
Università
di Catania



Machine Learning

Giovanni Maria Farinella

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http://www.dmi.unict.it/farinella/

FPV@IPLAB - http://iplab.dmi.unict.it/fpv

Lecture 1

- Docente e Tutor
- Struttura del Corso di ML
- Informazioni su esame e progetto
- Perchè ML è interessante?
- Applicazioni ed esempi
- ML @ DMI

www.dmi.unict.it/farinella

GIOVANNI MARIA FARINELLA

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Curriculum Vitae et Studiorum
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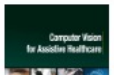
(mark fax ATTN. G.M. Farinella)

BOOKS

Computer Vision - ECCV 2022

Pattern Recognition. ICPR International Workshops and Challenges (link)

Book on CV for Assistive Healthcare (link)



Giovanni Maria Farinella obtained the degree in Computer Science (egregia cum laude) from the University of Catania, Italy, in 2004. He is Founder Member of the IPLAB Research Group at University of Catania since 2005. He also became an Associate Member of the Computer Vision and Robotics Research Group at University of Cambridge in 2006, and Associate Member of the Italian National Research Council since 2018. He was awarded a Doctor of Philosophy (Computer Vision) from the University of Catania in 2008. He is Full Professor at the Department of Mathematics and Computer Science, University of Catania, Italy. His research interests lie in the fields of Computer Vision, Pattern Recognition and Machine Learning, with focus on First Person (Egocentric) Vision. He is Associate Editor of the international journals IEEE Transactions on Pattern Analysis and Machine Intelligence, Pattern Recognition - Elsevier , International Journal of Computer Vision. He has been serving as Area Chair for CVPR 2020/21/22/23, ICCV 2017/19/21/23, ECCV 2020, BMVC 2020, WACV 2019/22, ICPR 2018, and as Program Chair of ECCV 2022, ICIAP 2021 and VISAPP 2019/20/21/22/23. Giovanni Maria Farinella founded (in 2006) and currently directs the International Computer Vision Summer School. He also founded (in 2014) and currently directs the Medical Imaging Summer School. He is member of the European Laboratory for Learning and Intelligent Systems (ELLIS), Senior Member of the IEEE Computer Society, Scientific Advisor of the NVIDIA AI Technology Centre (NVAITC), and board member of the CINI Laboratory of Artificial Intelligence and Intelligent Systems (lead of the area AI for Industry - since 2021). He was awarded the PAMI Mark Everingham Prize 2017 and the Intel's 2022 Outstanding Researcher Award. In addition to academic work, Giovanni's industrial experience includes scientific advisorship to different national and international companies and startups, as well as the leadership as Founder and Chief Scientific Officer of Next Vision - Spinoff of the University of Catania.

Scopus Profile, **Google Scholar Profile**

Giovanni Maria Farinella has got the National Scientific Qualification to Associate Professor (ASN - Seconda Fascia) in the areas of Information Engineering (SSD 09/H1) and Computer Science (SSD 01/B1) on 03/12/2013 and 29/01/2014 respectively.

He has got the National Scientific Qualification to Full Professor (ASN - Prima Fascia), in the areas of Computer Science (SSD 01/B1) and Information Engineering (SSD 09/H1) on 28/03/2018 and 26/07/2018 respectively.

***I*mage *P*rocessing *Lab*oratory**

First Person (Egocentric) Vision @ IPLAB

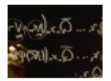
COURSES

SEMINARS & PRESENTATIONS

PUBLICATIONS

RESEARCH VISITS

PROJECTS


PROFESSIONAL ACTIVITIES

PICTURES

LINKS & EVENTS

NEWS

- **Are you looking for my help? Send me an email to fix a meeting. Do not hesitate!**
- **Research Scholarship Opportunities**

SCHOLARSHIPS

- **Ph.D. Opportunities**

PhD

- **PostDoctoral Opportunities**

POSTDOCTORAL

Rosario Leonardi

SEGUI

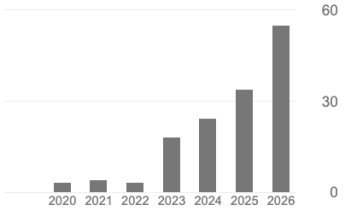
University of Catania
Email verificata su unict.it

Computer Vision Machine Learning Egocentric Vision

TITOLO	CITATA DA	ANNO
Exploiting multimodal synthetic data for egocentric human-object interaction detection in an industrial scenario R Leonardi, F Ragusa, A Furnari, GM Farinella Computer Vision and Image Understanding 242, 103984	39	2024
Enigma-51: Towards a fine-grained understanding of human behavior in industrial scenarios F Ragusa, R Leonardi, M Mazzamuto, C Bonanno, R Scavo, A Furnari, ... 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV …	38 *	2024
Egocentric Human-Object Interaction Detection Exploiting Synthetic Data R Leonardi, F Ragusa, A Furnari, GM Farinella Image Analysis and Processing–ICIAP 2022: 21st International Conference …	25	2022
Are Synthetic Data Useful for Egocentric Hand-Object Interaction Detection? R Leonardi, A Furnari, F Ragusa, GM Farinella European Conference on Computer Vision, 36-54	16	2025
Vedi: Vision exploitation for data interpretation GM Farinella, G Signorello, S Battiato, A Furnari, F Ragusa, R Leonardi, ... International Conference on Image Analysis and Processing, 753-763	15	2019
HERO: An Artificial Conversational Assistant to Support Humans in Industrial Scenarios C Bonanno, F Ragusa, R Leonardi, A Furnari, GM Farinella International Conference on Signal Processing and Multimedia Applications …	3	2022
ENIGMA-360: An Ego-Exo Dataset for Human Behavior Understanding in Industrial Scenarios F Ragusa, R Leonardi, M Mazzamuto, D Di Mauro, C Quattrocchi, ... arXiv preprint arXiv:2603.09741	2	2026
Leveraging Synthetic Data for Enhancing Egocentric Hand-Object Interaction Detection R Leonardi, A Furnari, F Ragusa, GM Farinella International Journal of Computer Vision 134 (6), 279	1	2026
EgoInteract: Synthetic Egocentric Videos Generation for Interaction Understanding and Anticipation R Leonardi, F Ragusa, D Materia, A Passanisi, J Fort, J Engel, ... arXiv preprint arXiv:2605.18214	1	2026
A Real-Time System for Egocentric Hand-Object Interaction Detection in Industrial Domains A Finocchiaro, AS Catinello, M Mazzamuto, R Leonardi, A Furnari, ... International Conference on Image Analysis and Processing, 457-469	1	2025
GlovEgo-HOI: Bridging the Synthetic-to-Real Gap for Industrial Egocentric Human-Object Interaction Detection A Spoto, R Leonardi, F Ragusa, GM Farinella arXiv preprint arXiv:2601.09528		2026
Learning Egocentric In-Hand Object Segmentation through Weak Supervision from Human		2025

Citata da

	Tutte	Dal 2021
Citazioni	141	138
Indice H	5	5
i10-index	5	5



Accesso pubblico

VISUALIZZA TUTTO

0 articoli — 3 articoli

Non disponibili — Disponibili

In base ai mandati di finanziamento

Coautori

- Giovanni Maria Farinella — University of Catania
- Francesco Ragusa — University of Catania - IPLab (Im…
- Antonino Furnari — Associate Professor at the Unive…

https://iplab.dmi.unict.it/live

LIVE @ IPLAB

ABOUT IPLAB

The Image Processing Laboratory (IPLAB) is part of the Department of Mathematics and Computer Science of the University of Catania, Italy. IPLAB's research focuses in the areas of Image Processing, Computer Vision, Machine Learning and Computer Graphics.

LEARN MORE ABOUT IPLAB

CONTACT PERSONS FOR RESEARCH ON FIRST PERSON (EGOCENTRIC) VISION

Prof. Giovanni Maria Farinella
giovanni.farinella@unict.it

HOME

LAST UPDATE: SEP 10, 2025

LIVE @ IPLAB

The terms *First Person Vision* or *Egocentric Vision* refer to the study and development of Computer Vision techniques in the scenario in which images and video are acquired from the user's point of view. This is generally done employing *wearable cameras* such as Google Glass, Microsoft HoloLens and GoPro. This acquisition paradigm is in contrast with standard *Third Person Vision* applications which assume that images are acquired by fixed cameras.

Visual content acquired according to the *First Person Vision* paradigm is inherently different from standard *Third Person Vision Content*. Indeed, while a fixed camera observes event from a *neutral* point of view, egocentric visual content captures the personal visual experience of the camera wearer. While Third Person Vision content is often edited and pre-segmented (e.g., movies or collections of YouTube videos), egocentric video is generally acquired in a continuous fashion and hence it tends to be unstructured and difficult to index. While strong assumptions can be generally made on Third Person Visual content, First Person data is inherently characterized by a continuously changing context which must be dealt with.

With its intrinsic mobility, *First Person Vision* poses some new challenges (e.g., changing context, motion blur, unstructured content), and offers unique opportunity to develop true intelligent systems able to assist the user and augment his abilities.

The aim of this page is to present research done by the LIVE Group of the Image Processing LABoratory (IPLAB) in the scope of First Person Vision.

LIVE

LEARNING | INTELLIGENCE | VISION | EMBODIEMENT

Struttura del corso

9 Lezioni Teoriche

9 Laboratori

2 Lezioni Industriali: Next Vision, Xenia

2 Lezioni di Approfondimento (Data Law) /Ricerca (Medical Imaging)

2 Prove in Itinere: 16 Dicembre 2026, 11 Gennario 2026

Sito e Materiale Didattico: http://www.dmi.unict.it/farinella/ML

- Credenziali
  - studente
  - corsoml

Canale Teams – Codice: j8rls3j

Libri: la biblioteca del DMI è provvista di libri utili per approfondimenti. Si veda anche il sito del corso per riferimenti.

Argomenti:
Concetti Base
Linear/Logistic Regression
Polinomial Linear/Logistic Regression
Perceptron (Xor Problem)
SoftMax e Regularization
Decision trees
Random Forests
Kernel Machines
Combining Methods

Modalità di Esame

- Esami
  - Scritto + Progetto (in team max 3 persone)
  - 50% Scritto, 50% Progetto
- Maggiori dettagli sulla modalità di svolgimento dei progetti e la consegna delle relazioni verranno date alla prima lezione di lavoratorio

Stage e Tesi

- Sperimentali
- Anche collaborazione con Aziende (Next Vision, Xenia, STMicroelectronics, ecc..) e Enti di Ricerca
- E’ consigliato iniziare la tesi in tempo (6+ mesi)

Why is interesting to study ML

Application Domains and Examples

ML Application Domains

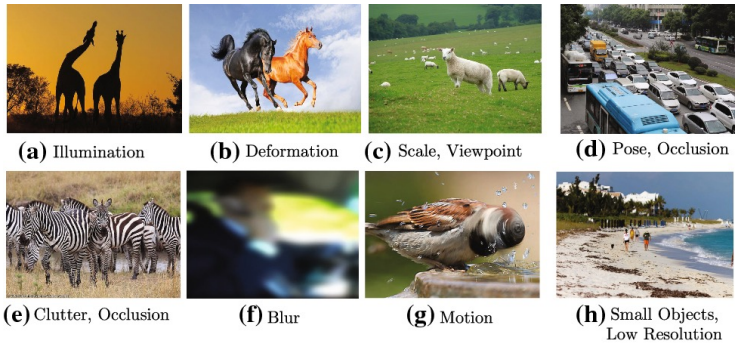
- Computer Vision
- Robotics
- Gaming
- Natural Language Processing and Speech
- Art
- …

Machine Learning for Computer Vision

- Object Recognition and Segmentation (guess the input and the output)
- Image Captioning
- Human pose Estimation
- Summarization and Indexing
- Localization
- First Person Vision for Anticipation

Why is interesting

- Vision is challenging!
  - For a small patch of resolution 256x256 with 256 pixel values
  - A total of $2^{8x256x256} = 2^{524288}$ of possible images
  - In comparison there are about ~10^{24} stars in the universe
- Image variability
  - Different viewpoint, scales, deformations, occlusions…
- Semantic Variation
  - Big Intra-class Variation
  - Small Inter-class Variation



(i) Different instances of the "chair" category

(j) Small Interclass Variations: four different categories

Machine Learning for Robotics / Artificial Agents

- Self-Driving Cars (guess the input and the output)
- Autonomous Robots
- Activity Recognition
- Tracking and Location Estimation
- Gaming

Why is interesting

- Agents (e.g., Robots) are usually considered in controlled environments
  - Laboratory settings, Predictable positions, Standardized tasks (like in factory robots)
- Real Life
  - Environments constantly change, new tasks need to be learnt without guidance, unexpected factors must be dealt with

Machine Learning for Natural Language Processing, Speech Recognition, Audio Recognition

- Language Understanding
- Speech Understanding
- Audio recognition
- Conversational Agents (e.g., ChatGPT)

Why is interesting

- NLP is a complex task
  - synonymy (“chair”, “stool” or “beautiful”, “handsome”)
  - ambiguity (“Cut to the chase”)
- NLP is very high dimensional
  - assuming 150K English words, we need to learn 150K classifiers
  - with quite sparse data for most of them

Machine Learning for Art and Computer Graphics

- Imitating Famous Painters
- Image to Image Translation
- Generating Music
- Generating Films
- Generating Realistic Images
- Generating Avatars
- Generating Video

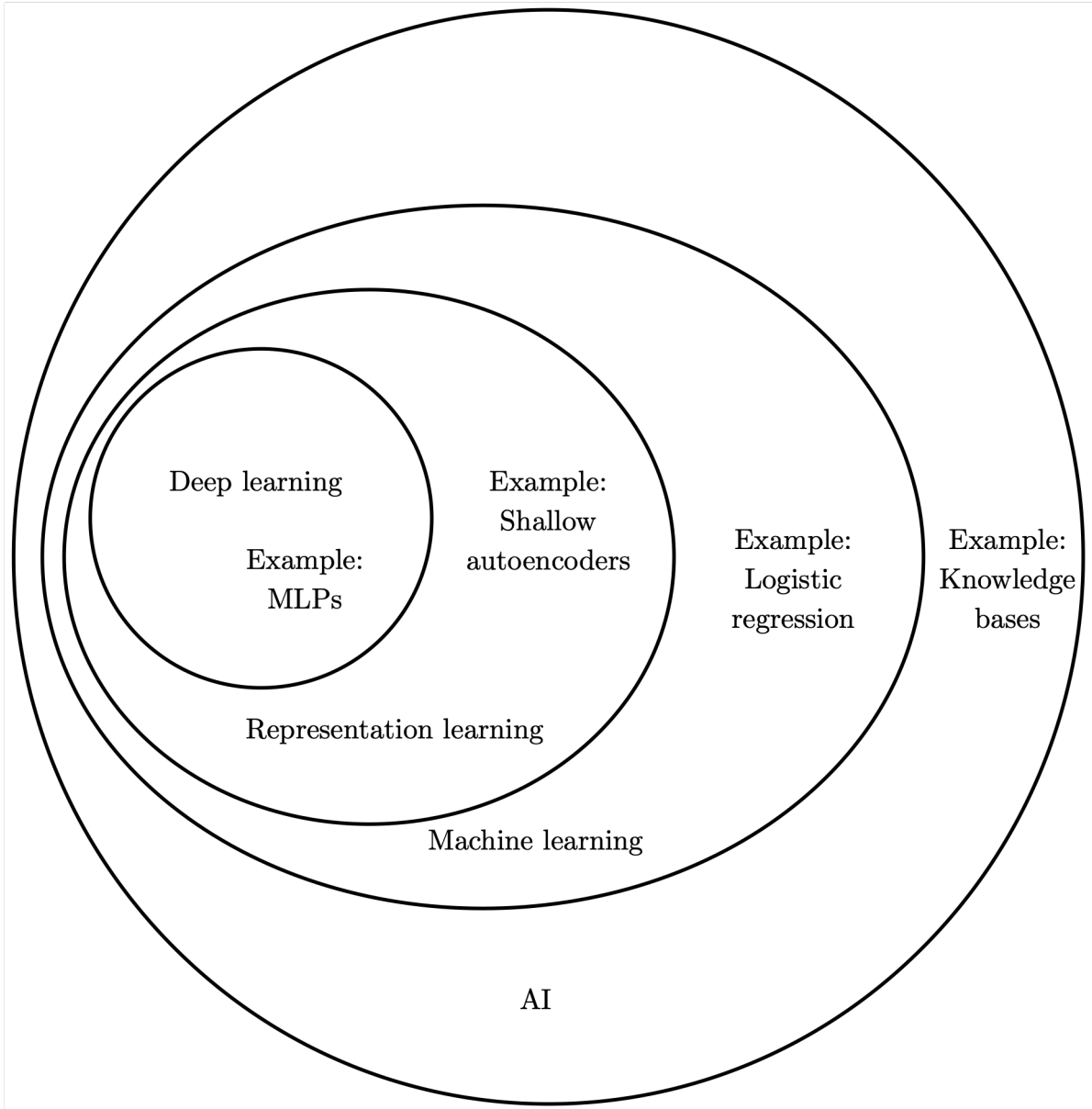
Why is interesting

- Music, painting, etc. are tasks that are uniquely human
  - Difficult to model
  - Even more difficult to evaluate (if not impossible)
- If machines can generate novel pieces even remotely resembling art, they must have understood something about “beauty”, “harmony”, etc.
- Have they really learned to generate new art, however?
  - Or do they just fool us with their tricks?

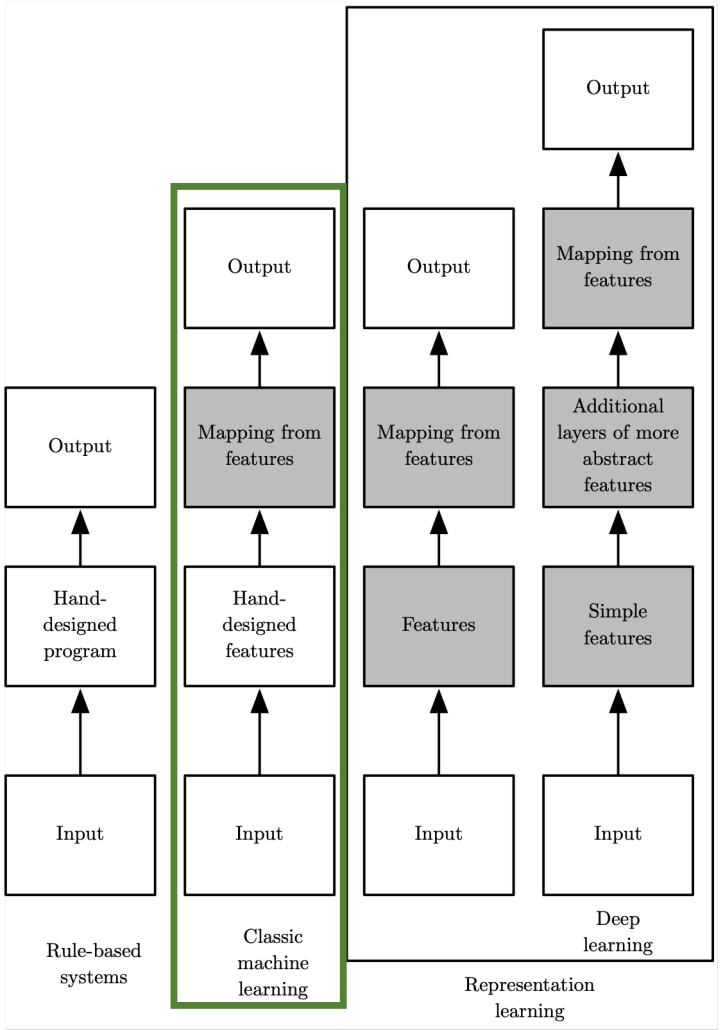
We can continue giving examples on security, chemistry, industry, etc, etc… but you have got the gist!

Nowadays, ML is basically everywhere

The basis of ML are fundamental to build intelligent machines with modern AI technologies based on Deep Learning


Deep learning
Example:
MLPs
Example:
Shallow
autoencoders
Representation learning
Example:
Logistic
regression
Machine learning
Example:
Knowledge
bases
AI

What this course is about



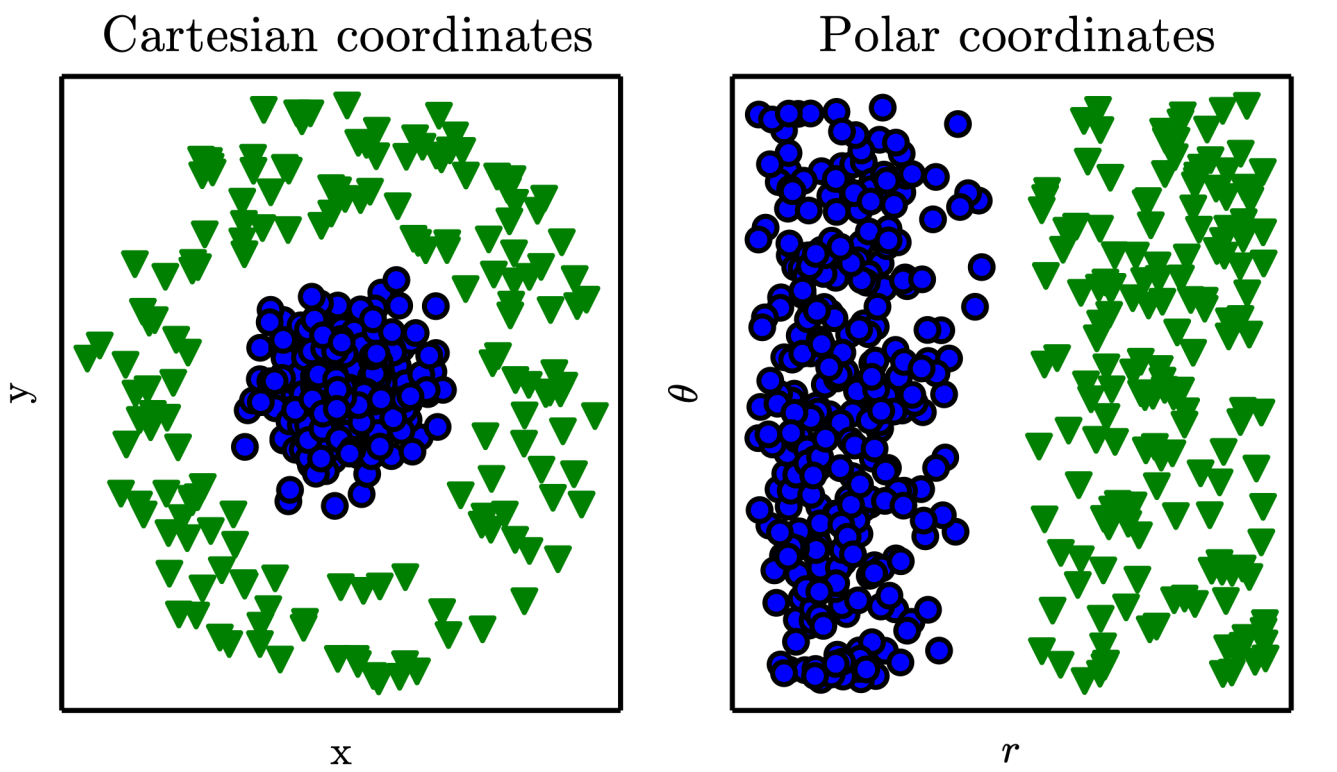
Learning Procedure

Deep Learning Book (www.deeplearningbook.org)

Representation of Data Matter

Two modalities:
- Hand Designed Representations
- Learned Representations

- **Blue class**: points inside a circle centered at the origin.
- **Green class**: points outside that circle.
- In Cartesian coordinates (x, y), the classes are **not linearly separable**.
- In polar coordinates (r, θ), the classes become linearly separable because the decision depends only on the radius r.



Deep Learning Book (www.deeplearningbook.org)

Representation of Data Matter

Cartesian Coordinate

Let

$$r = \sqrt{x^2 + y^2}.$$

Assume the separating circle has radius R.

The optimal classifier is

$$f(x, y) = x^2 + y^2 - R^2.$$

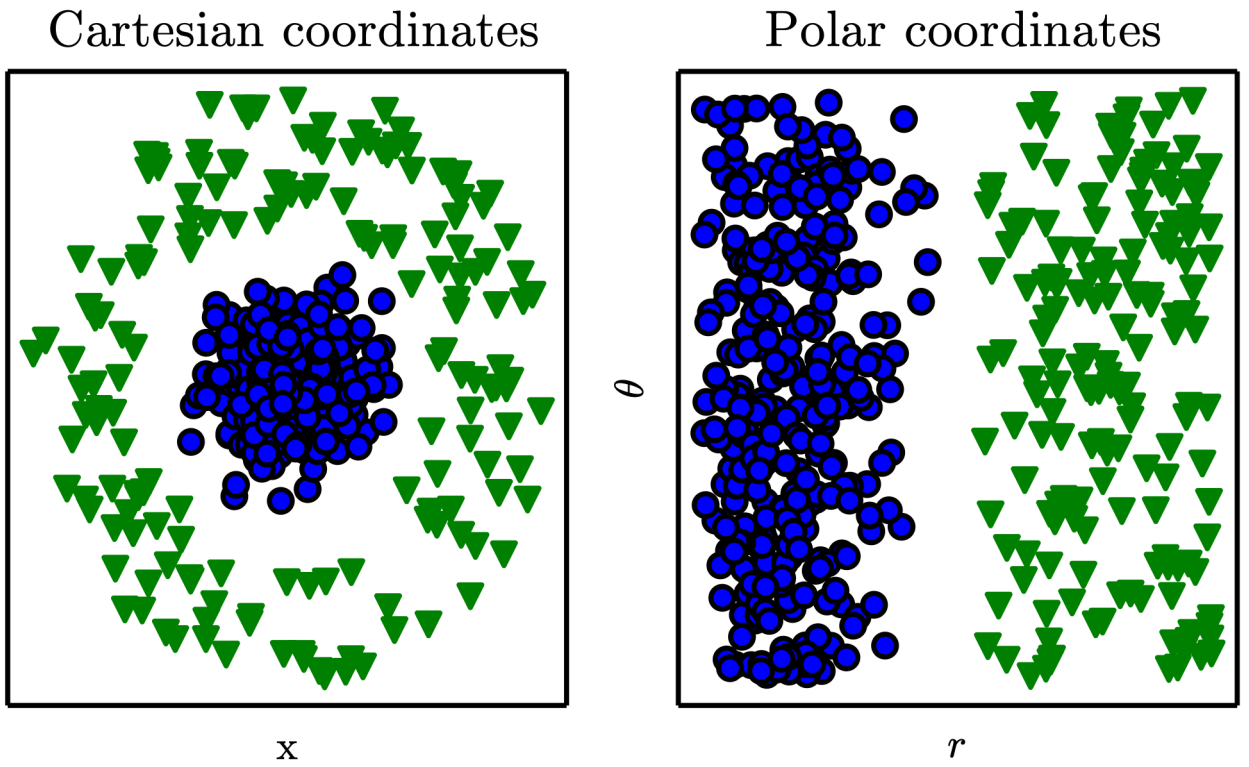
Decision rule:

$$\hat{y} = \begin{cases} \text{Blue} & \text{if } x^2 + y^2 < R^2, \\ \text{Green} & \text{if } x^2 + y^2 > R^2. \end{cases}$$

The decision boundary is

$$x^2 + y^2 = R^2,$$

which is a circle.



Deep Learning Book (www.deeplearningbook.org)

Representation of Data Matter

Polar Coordinate

$$f(r, \theta) = w_r r + w_\theta \theta + b,$$

then the optimal solution is simply obtained by making the classifier independent of θ:

$$w_\theta = 0.$$

Hence,

$$f(r, \theta) = w_r r + b.$$

If the circle separating the two classes has radius R, choose

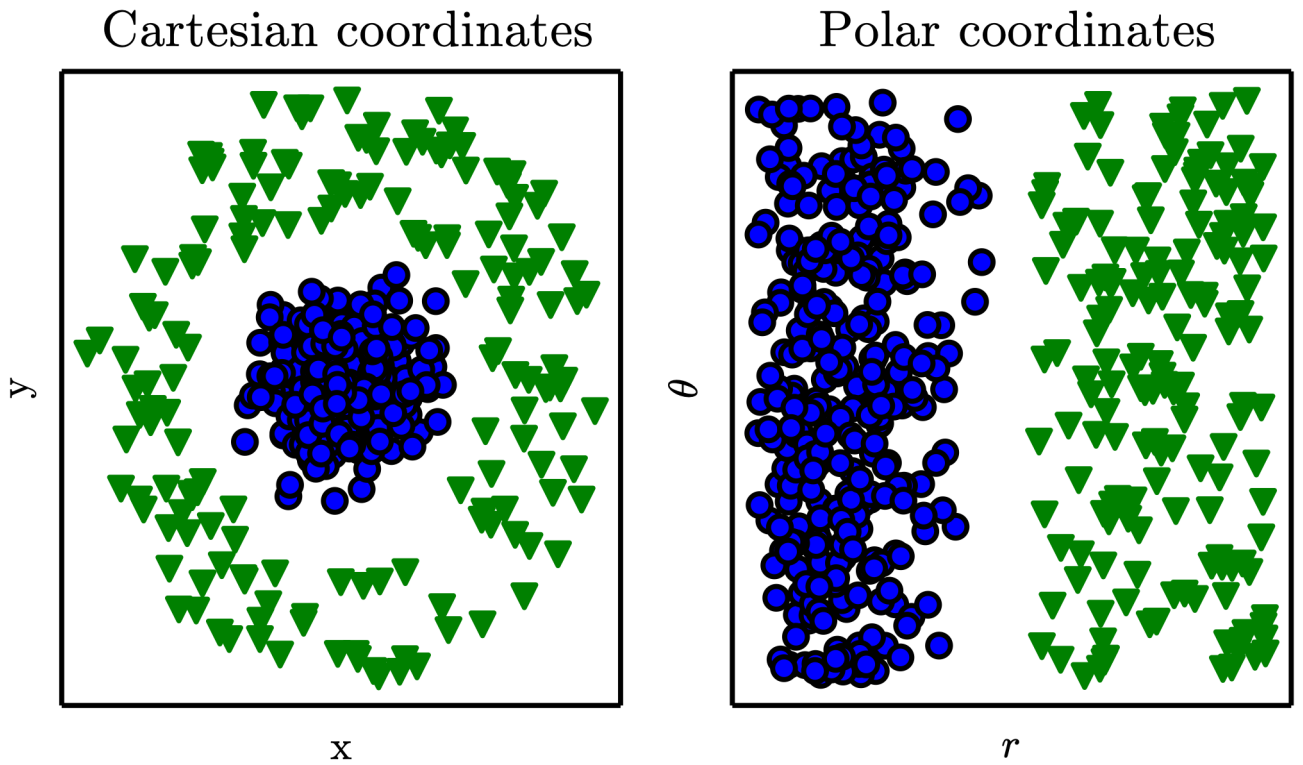
$$w_r = 1, \qquad b = -R.$$

The decision function becomes

$$\boxed{f(r, \theta) = r - R}$$

and the classification rule is

$$\hat{y} = \begin{cases} +1 & \text{if } r > R, \\ -1 & \text{if } r < R. \end{cases}$$



How to choose the representation?
How to choose the model?
How to find parameters?
How to evaluate the model?

Deep Learning Book (www.deeplearningbook.org)

Main ML Ingredients

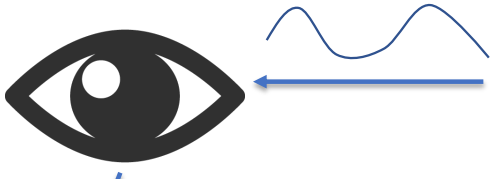
- Tasks: the problem to be solved (e.g., Classification, Image Generation, …)
- Data: examples Input-Output
- Algorithms: mathematical models (parametrized functions) to learn from data
- Hardware: CPUs vs GPUs
- Evaluation Protocols: to measure performances
- Humans: to design and develop ML, as well as to curate data to be used for training ML

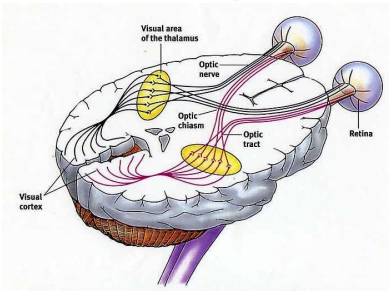
ML @ DMI

Visual Intelligence

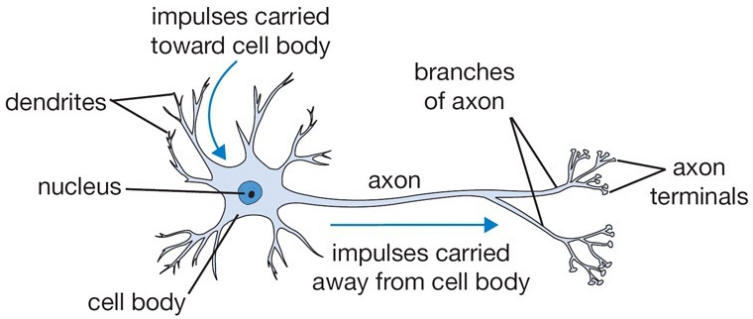
“learning to answer questions about images and videos”

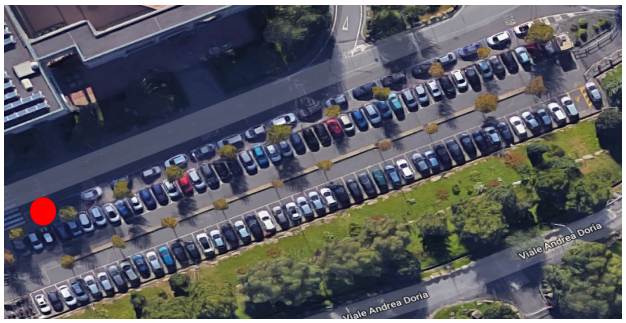
Human Visual System





Human vision involves over 60 billion neurons.





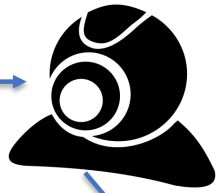
T A S K

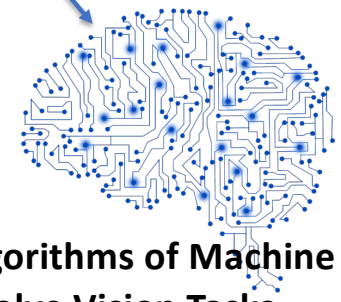
How many cars are in the parking area?

103

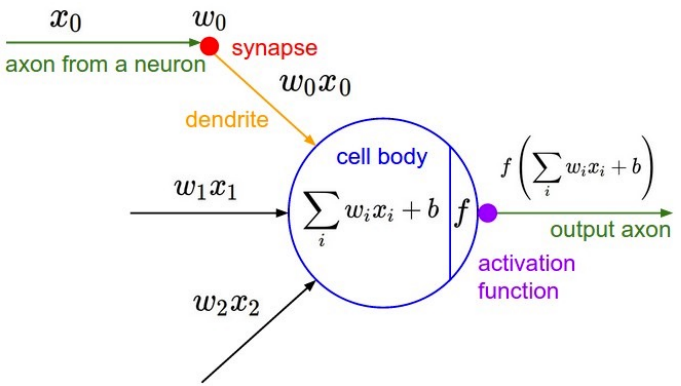
Visual Artificial System

101011001

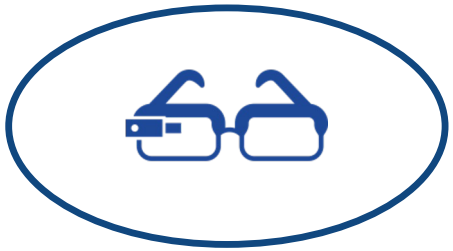




Advanced Algorithms of Machine Learning to Solve Vision Tasks

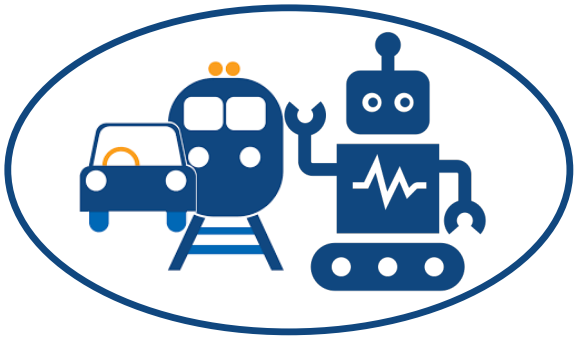


Visual Intelligence - “Points of Views”



First Person Vision - Point of View of the user

Egocentric Vision



Embodied Vision

Egocentric Vision

Third Person

Exocentric Vision

www.dmi.unict.it/farinella - gfarinella@dmi.unict.it

Egocentric vision or **first-person vision** is a sub-field of computer vision that entails **analyzing images, videos captured by a wearable device**, which is typically worn on the head or on the chest and **naturally approximates the visual field of the camera wearer**.

Consequently, **visual data capture the part of the scene on which the user focuses to carry out the task** at hand and offer a valuable perspective **to understand the user's intents and activities and their context in a naturalistic setting.**

This research area is sometimes referred with the name "Wearable Vision".

Egocentric vs Exocentric

Exocentric

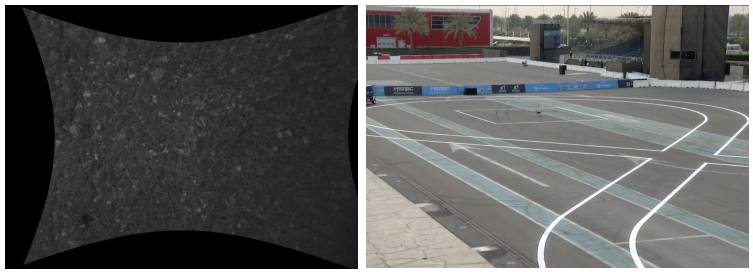
Egocentric

ENIGMA-51: Towards a Fine-Grained Understanding of Human Behavior in Industrial Scenarios. F. Ragusa R. Leonardi, M. Mazzamuto, C. Bonanno, R. Scavo, A. Furnari, G. M. Farinella. WACV (2024).

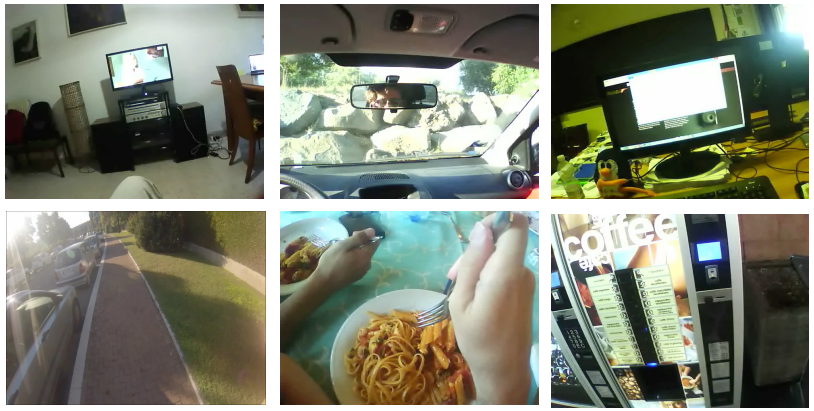
Egocentric (First-Person) Vision

Autonomous — The Human Role and the Point of View — Human

Egocentric Vision for Autonomous Robots



Egocentric Vision for Market Basket Analysis



Egocentric Vision to Support the User

- Stress Monitoring and Memory Augmentation
- Summarization and Lifelogging
- Personalized Imaging and Personal Big Data
- Assistive and Quality of Life Applications
- Virtual Assistant
- Object Interaction Prediction
- Action Anticipation
- Behavioural Analysis

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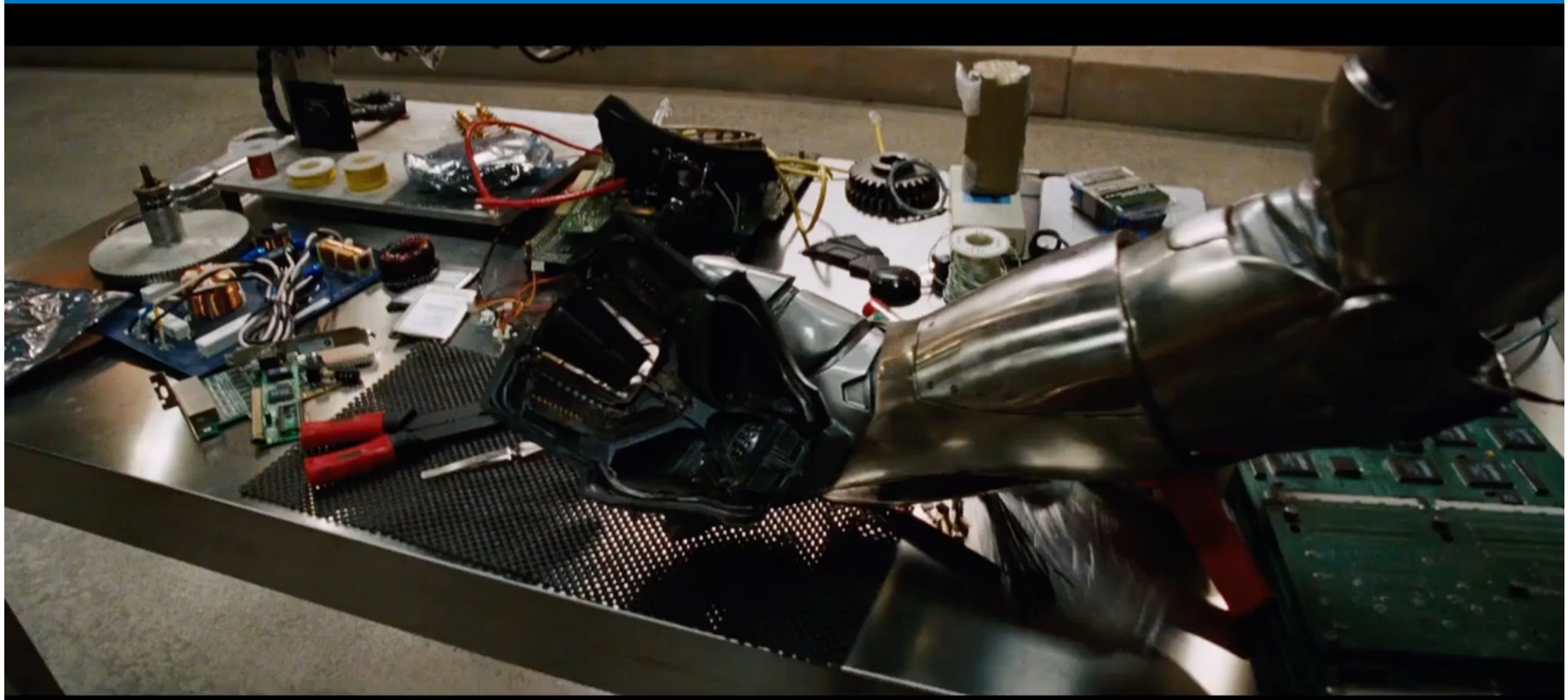
Università di Catania

What will the future AI be? We wonder!



Imagine to wear an AI-powered device able to observe the scene from your point of view to understand what you are doing and to support you in every day life where you live and work

An AI-Powered Wearable Virtual Assistant

https://thenounproject.com/Turkkub/

A wearable device which perceives the world from our "egocentric" point of view is ideal for implementing an AI-Powered Virtual Assistant

Examples of Device in the Market

First Person (Egocentric) Vision Devices

OrCam MyEye
Since 2015
Health, Assistive Technologies

Microsoft HoloLens
Since 2016
Industry, Mixed Reality

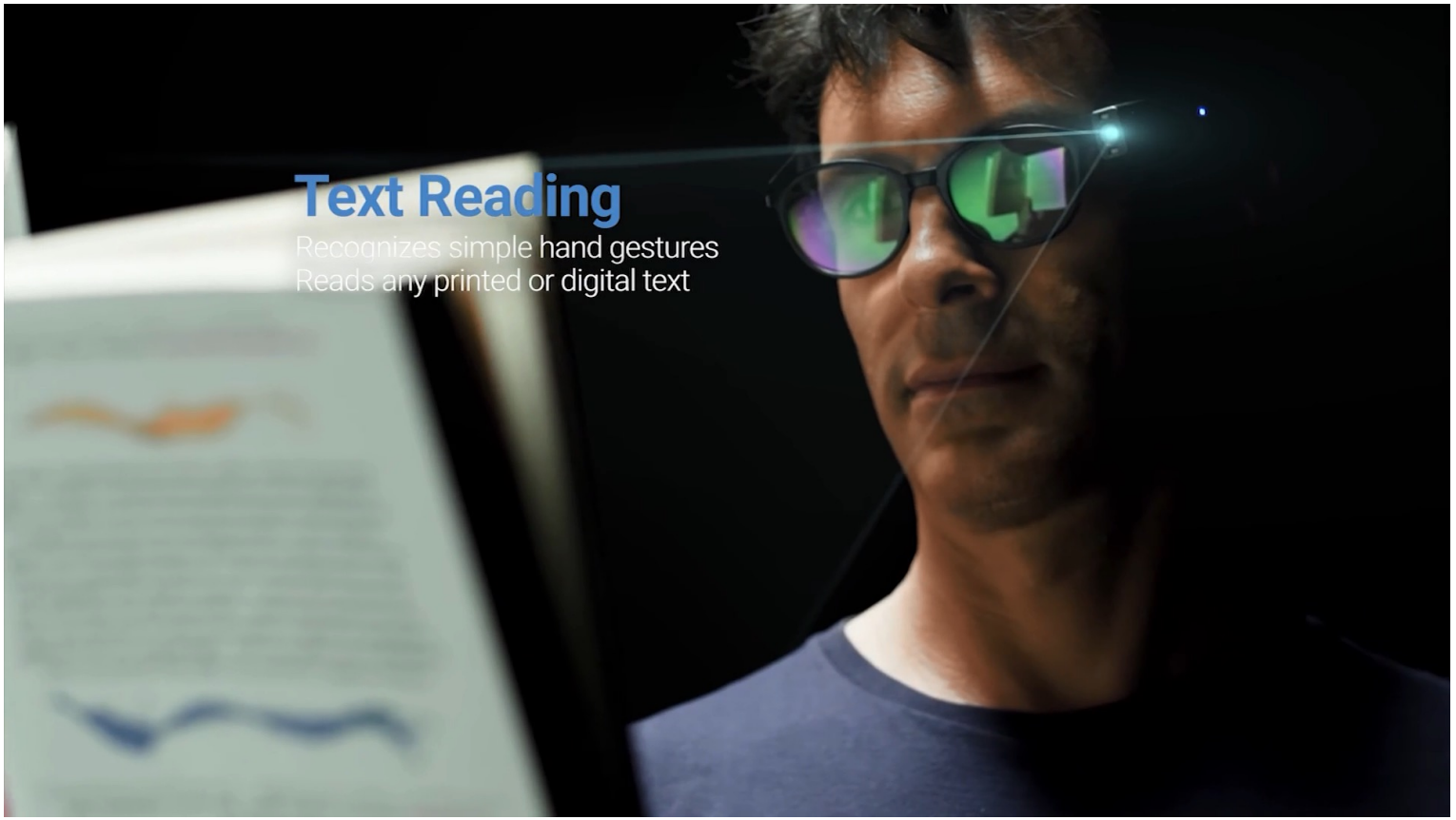
Apple Vision Pro
Since 2024
Health, Mixed Reality

OrCam MyEye, since 2015

Health, assistive technologies

https://www.orcam.com/

OrCam MyEye, since 2015



https://www.orcam.com/

You’ll want to put down your phone for this

Project ARIA

Meta

Egocentric Vision Research – Before EPIC

Egocentric Vision has a long research history…

Some goals of my research goup

- Video Understanding, Egocentric Vision, Embodied Intelligence
  - Algorithms to understand users' intent and actions performed with hands and feet
  - Procedural Understanding: Temporal Action Segmentation, Action Anticipation, human-object Interactions, Skill Assessment, Mistake Detection
  - Recognition of the objects observed and by the users
  - Localization of the agent and recognition of environment
  - Exploiting Synthetic Data for Learning
  - Formalise representations that can be useful for long-term understanding
  - Knowledge Transfer
- Application Domains – Binding with Industry
  - Home Assisted Living
  - Retail
  - Cultural Heritage
  - Industrial Environments
- Supporting benchmarks → need to push the community working in the field
  - few common goals and tasks, no common platform/standard/datasets

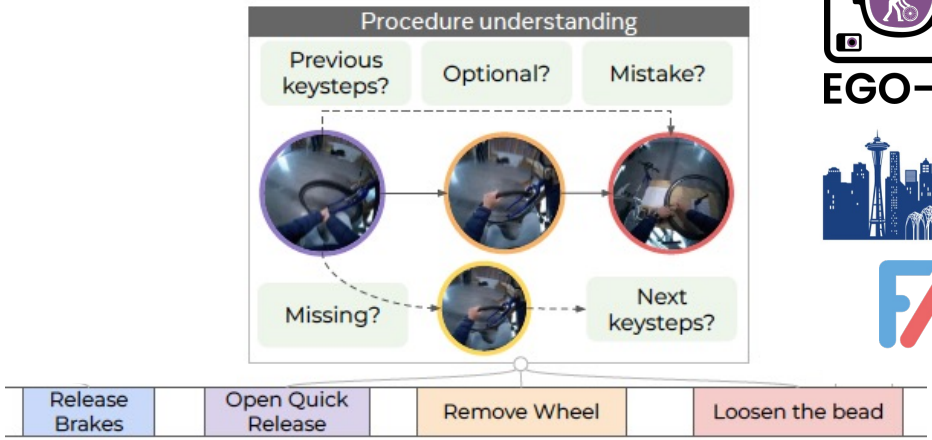
www.dmi.unict.it/farinella - gfarinella@dmi.unict.it

For what Wearable AI Assistants can be useful in real life?

Among the others, they can support humans in procedural tasks with the aim of improving health and safety in workplaces!

Procedure Understanding from Egocentric Vision

Procedural activities are sequences of key-steps aimed at achieving specific goals.



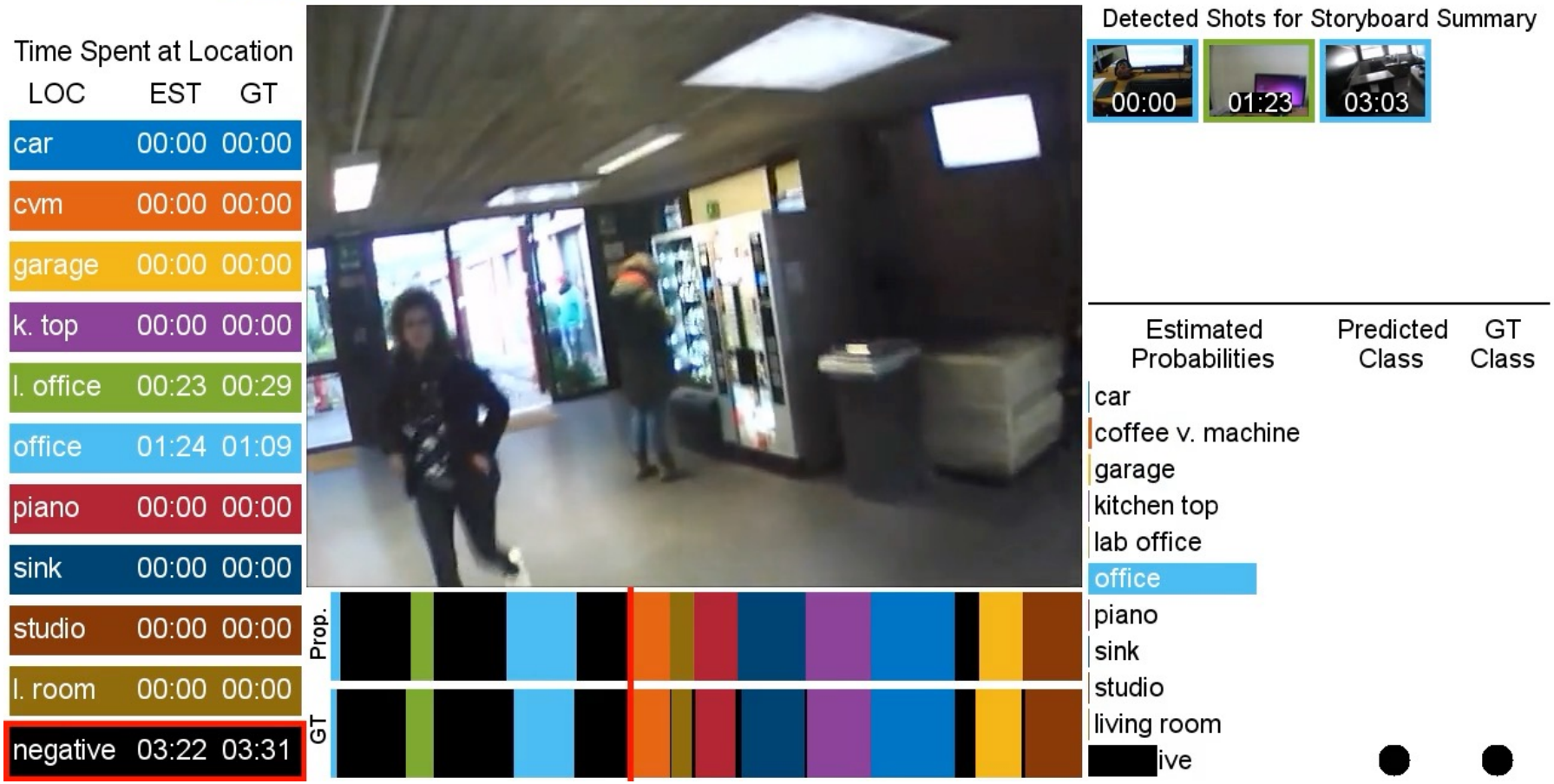
We wish to have AI that given a video segment s_t and its previous video segment history, AI models have to: 1) determine previous keysteps (to be performed before s_t); infer if s_t is 2) optional or 3) a procedural mistake; 4) predict missing keysteps (should have been performed before s_t but were not); and 5) next keysteps (for which dependencies are satisfied).

Grauman et al. (2024). Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. In Conference on Computer Vision and Pattern Recognition (CVPR)

What Wearable AI must be able to do to understand procedures?

What we did so far at UNICT and Next Vision to reach procedural understanding goals?

Procedure Understanding - Recognizing Context

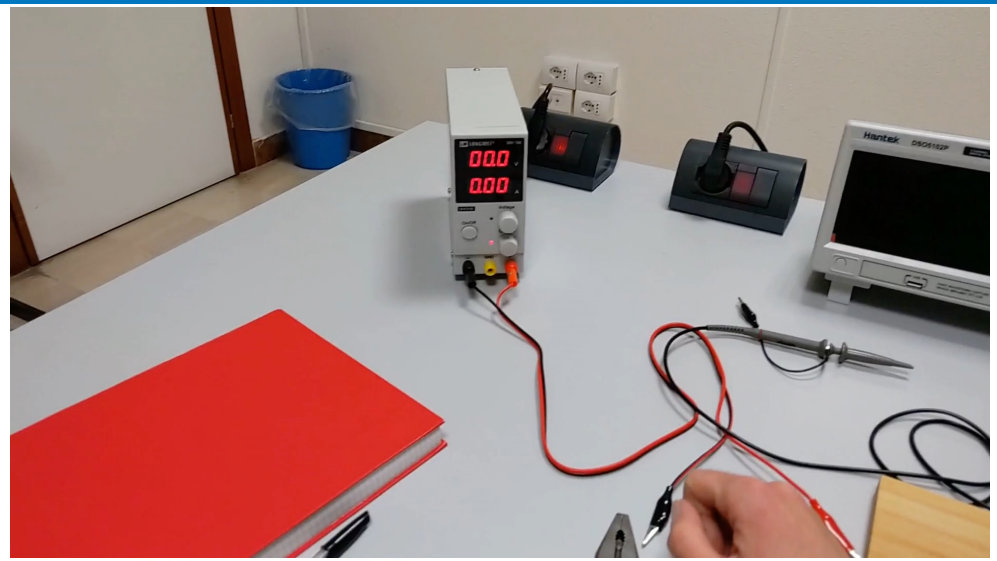


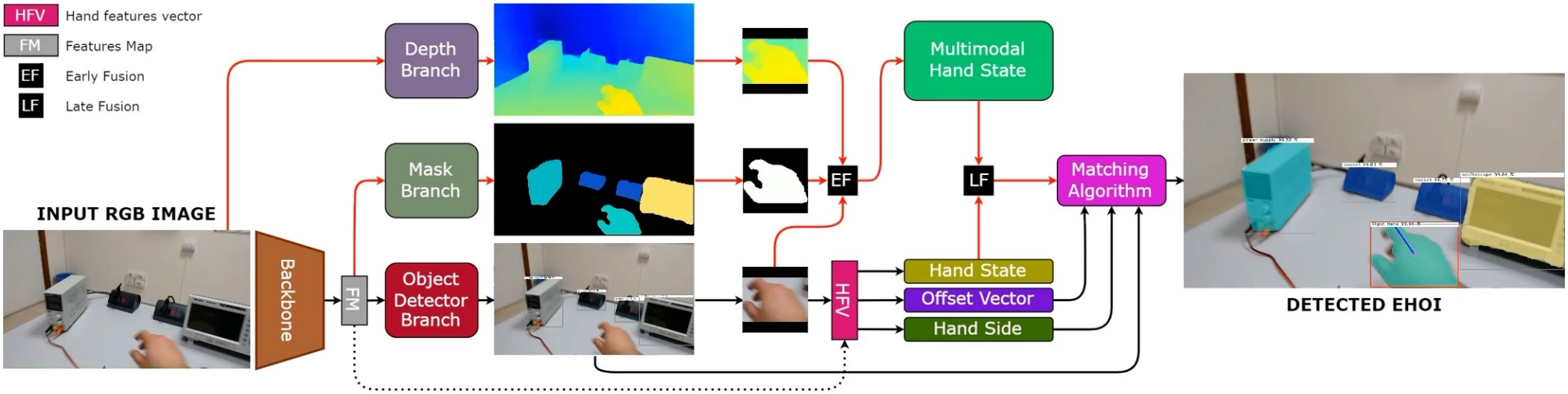
Antonino Furnari, Giovanni Maria Farinella, Sebastiano Battiato (2015). Recognizing Personal Contexts from Egocentric Images. In Workshop on Assistive Computer Vision and Robotics (ACVR) in conjunction with ICCV, Santiago, Chile, December 12

Antonino Furnari, Sebastiano Battiato, Giovanni Maria Farinella (2018). Personal-Location-Based Temporal Segmentation of Egocentric Video for Lifelogging Applications. Journal of Visual Communication and Image Representation , 52 , pp. 1-12

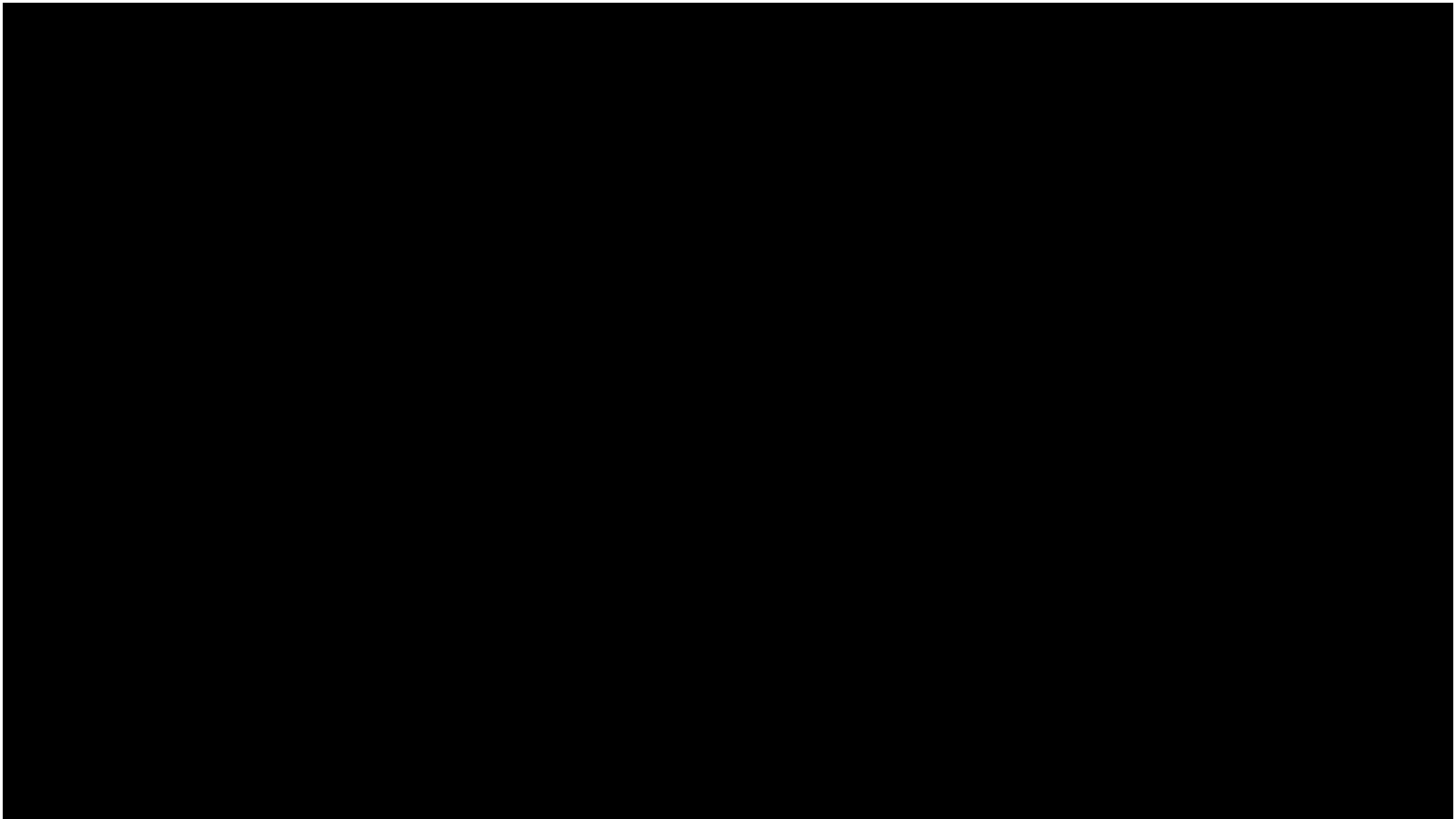
Procedure Understanding – Localization and Navigation

Procedure Understanding – Human Object Interaction

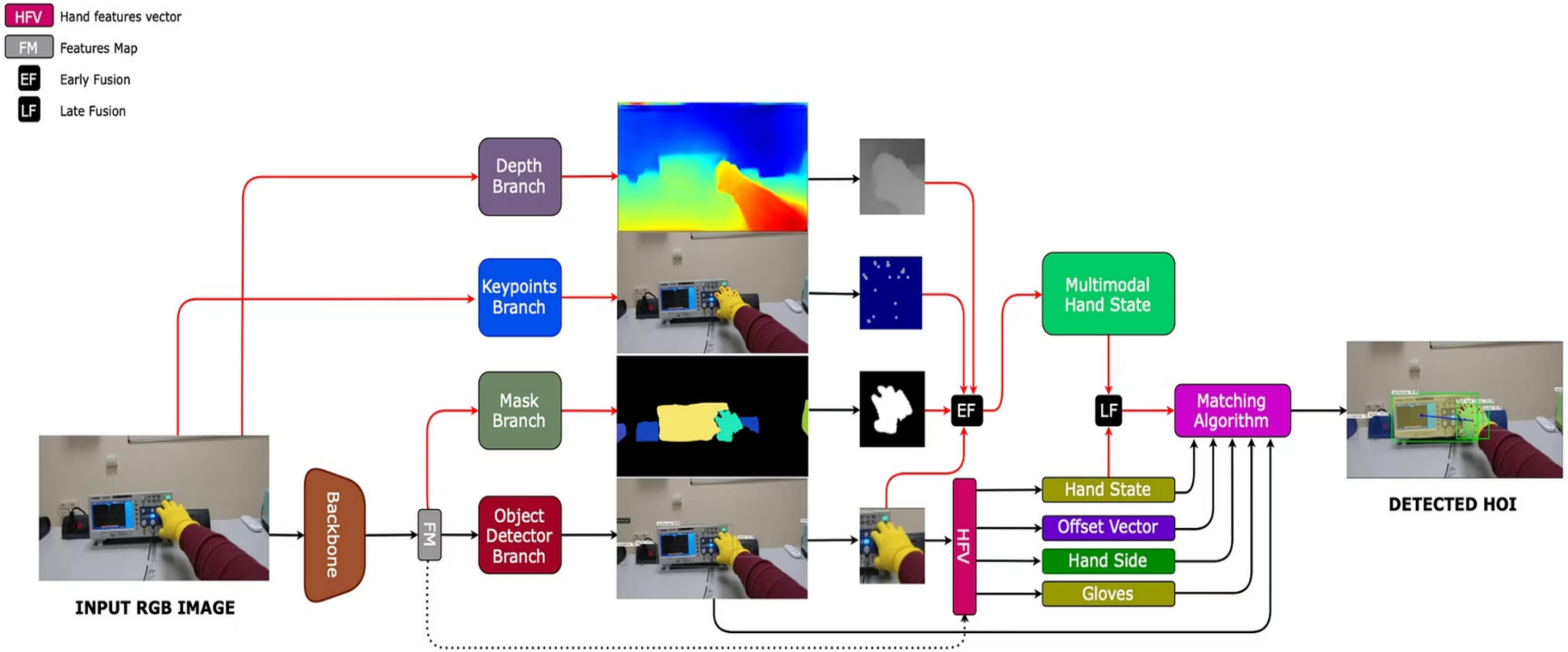




Rosario Leonardi, Francesco Ragusa, Antonino Furnari, Giovanni Maria Farinella (2024). Exploiting Multimodal Synthetic Data for Egocentric Human-Object Interaction Detection in an Industrial Scenario . Computer Vision and Image Understanding (CVIU)



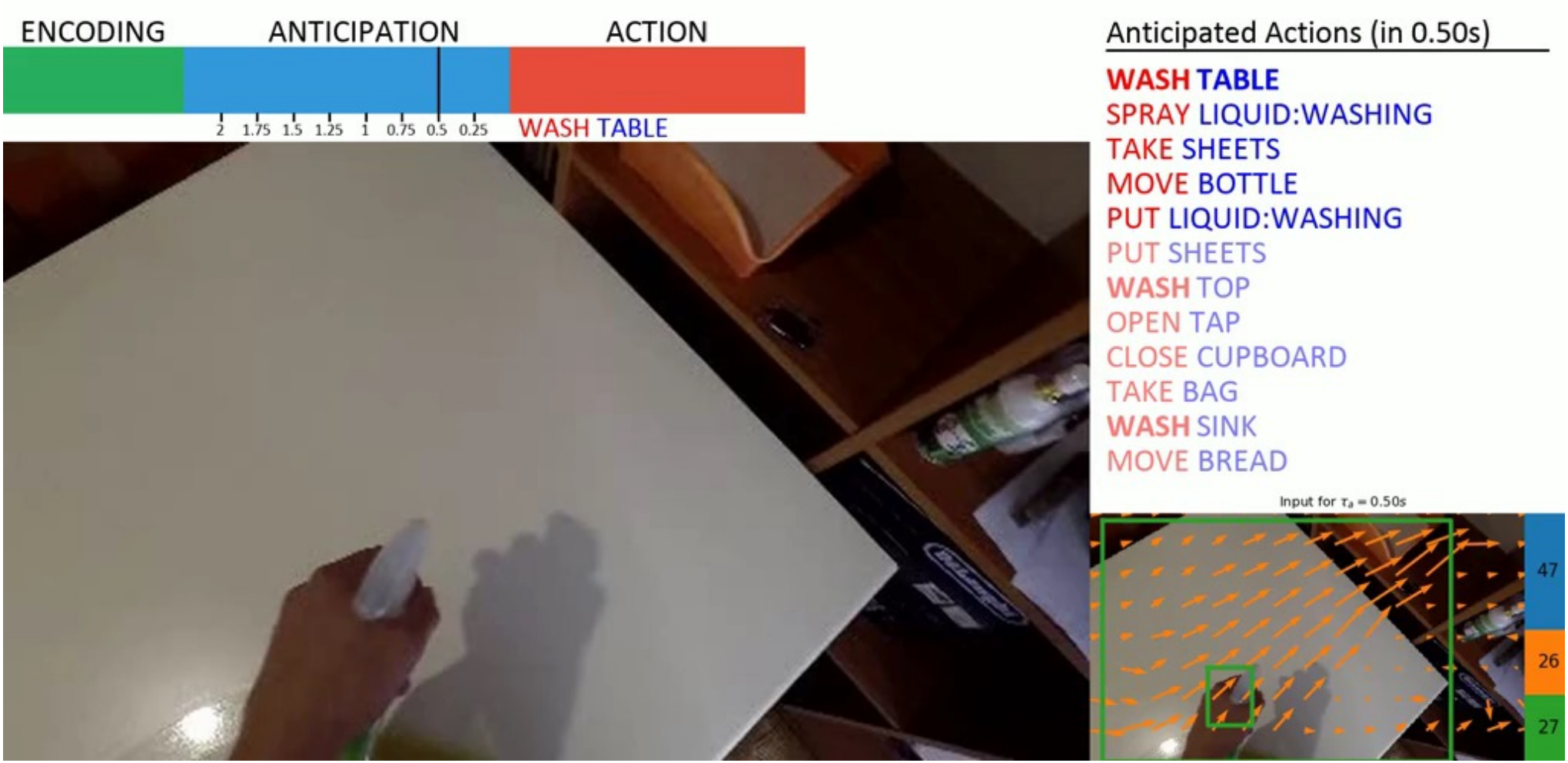
GLOVEGO-NET ARCHITECTURE



Output: ***<hand, contact state, active object, glove>***

Alfio Spoto, Rosario Leonardi, Francesco Ragusa, Giovanni Maria Farinella (2026). GlovEgo-HOI: Bridging the Synthetic-to-Real Gap for Industrial Egocentric Human-Object Interaction Detection. In International Conference on Computer Vision Theory and Applications (VISAPP)

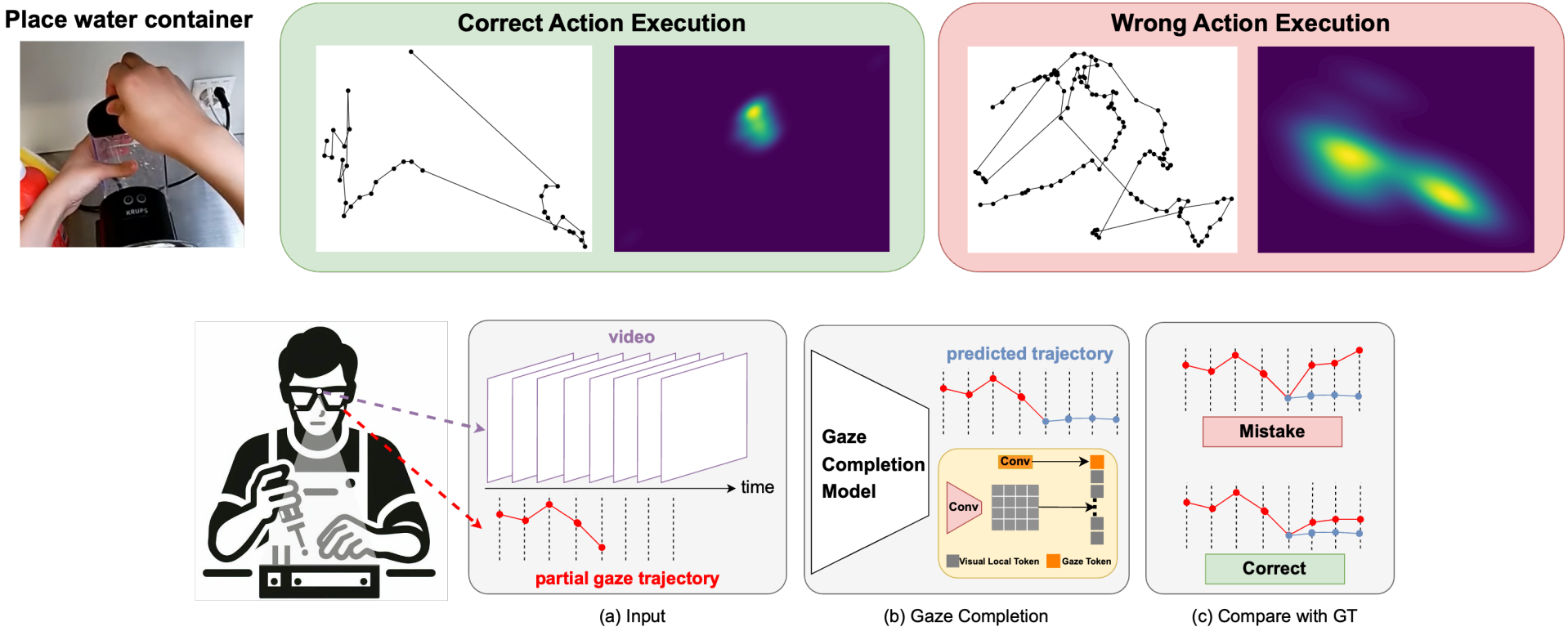
Procedure Understanding - Action Anticipation



A. Furnari, G. M. Farinella, What Would You Expect? Anticipating Egocentric Actions with Rolling-Unrolling LSTMs and Modality Attention. ICCV 2019 (ORAL)

A. Furnari, G. M. Farinella. Rolling-Unrolling LSTMs for Action Anticipation from First-Person Video. TPAMI 2021. http://iplab.dmi.unict.it/rulstm

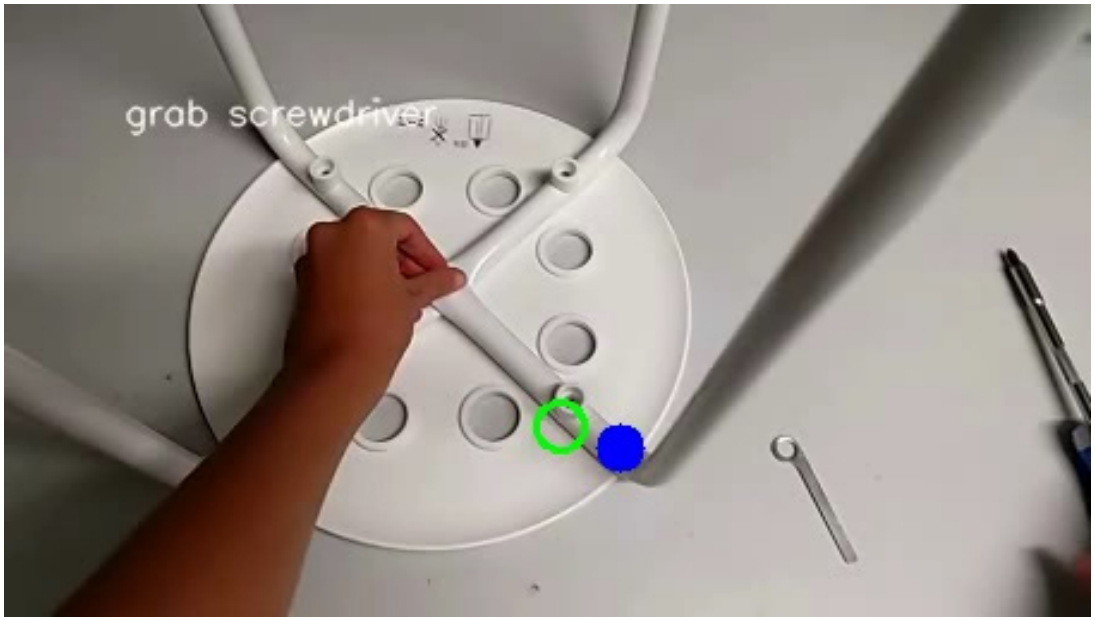
Procedure Understanding – Mistake Detection



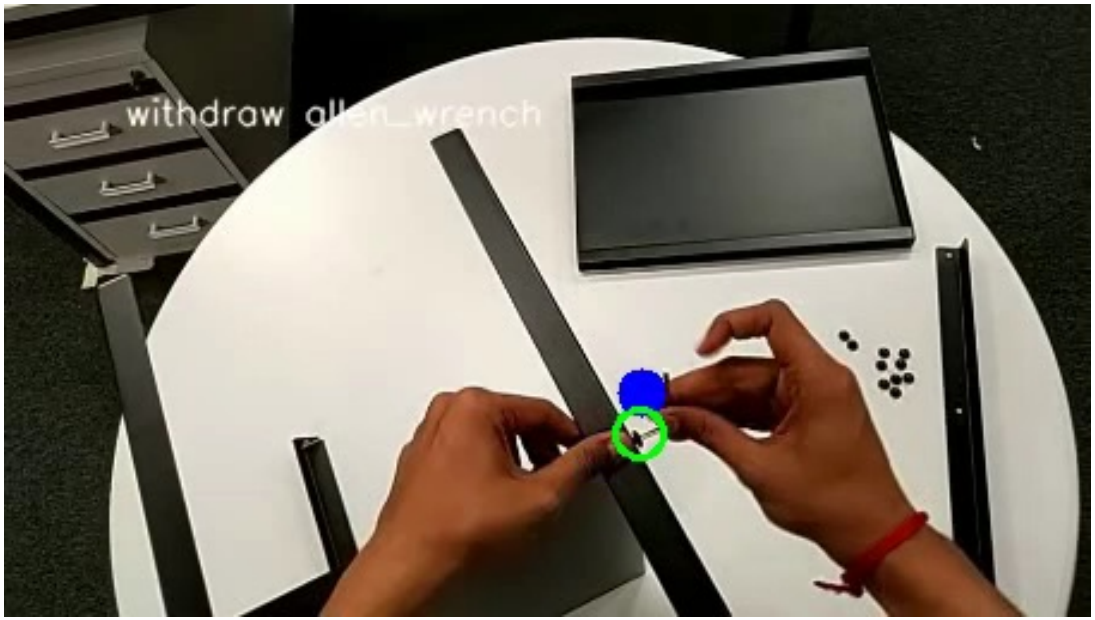
M. Mazzamuto, A. Furnari, Y. Sato, G.M. Farinella: "Gazing Into Missteps: Leveraging Eye-Gaze for Unsupervised Mistake Detection in Egocentric Videos of Skilled Human Activities by Detecting Unpredictable Gaze." CVPR, 2025.

Procedure Understanding – Mistake Detection

Correct



Mistake



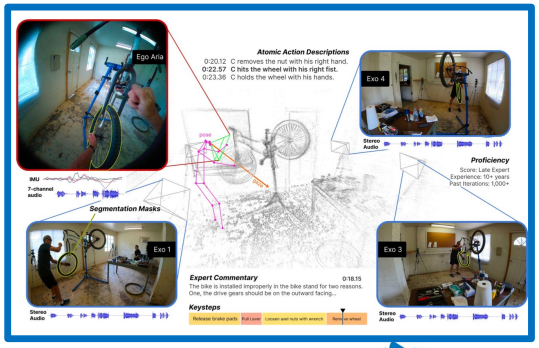
M. Mazzamuto, A. Furnari, Y. Sato, G.M. Farinella: "Gazing Into Missteps: Leveraging Eye-Gaze for Unsupervised Mistake Detection in Egocentric Videos of Skilled Human Activities by Detecting Unpredictable Gaze." CVPR, 2025.

Università di Catania — The Role of Real Data

MECCANO

The LIVE Research Group of UNICT has contributed to create different datasets for procedure understanding

Ego4D

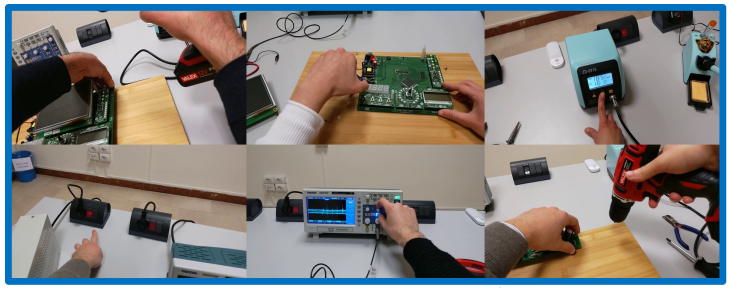


Ego-Exo4D

EPIC-KITCHENS

LIVE

LEARNING | INTELLIGENCE | VISION | EMBODIEMENT

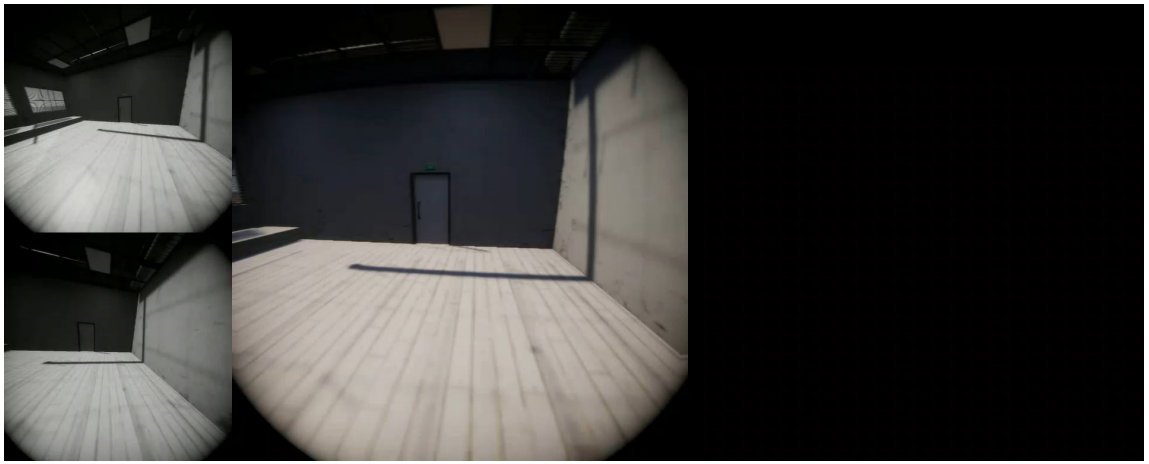


ENIGMA-51

HoloAssist

Assembly101

The Role of Syntetic Data



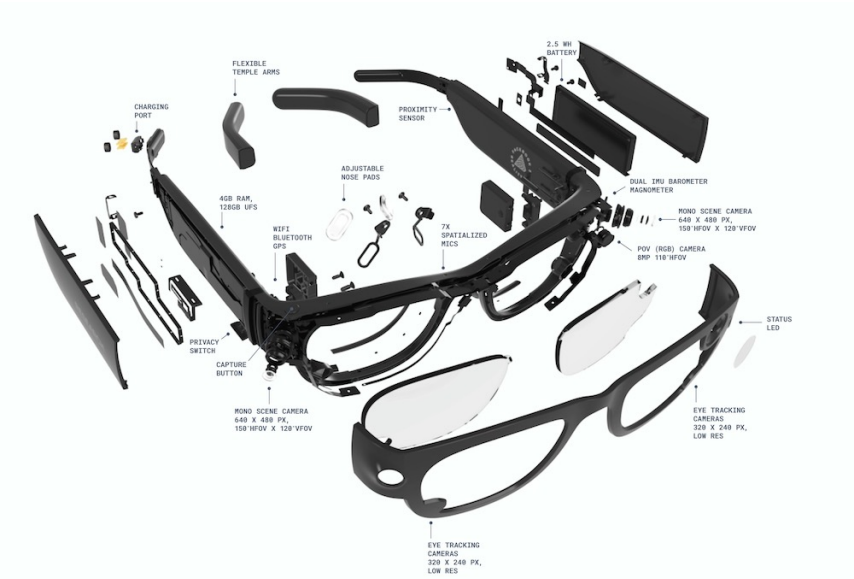
LIVE

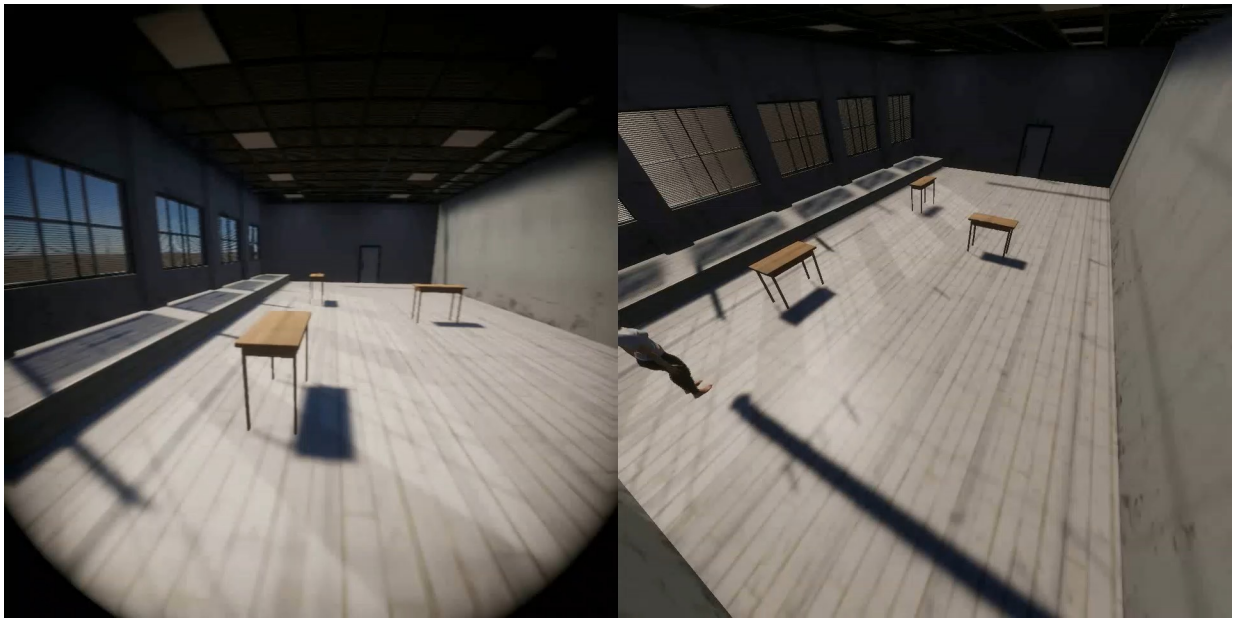
LEARNING | INTELLIGENCE | VISION | EMBODIEMENT

Meta

NEXT VISION

Spin-off of the University of Catania



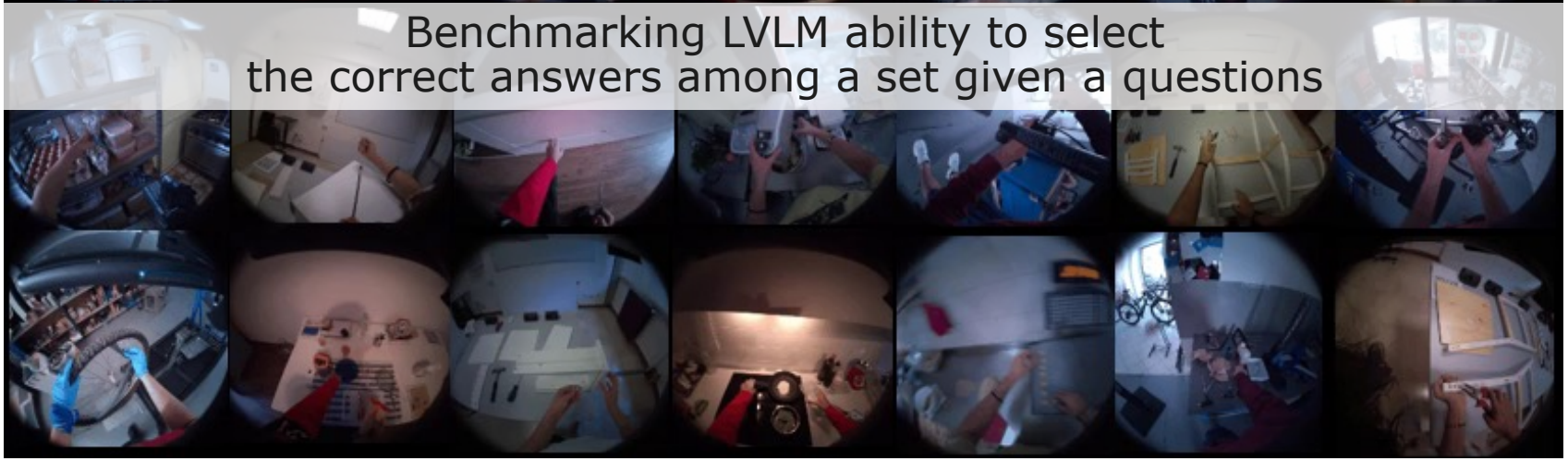


Are Large Visual-Language Models performing at the same level of humans in assisting operators during procedural tasks?

Are LVLM able to answer correctly to questions performed by an operator during procedures?

Università di Catania

Benchmarking LVLM ability to select
the correct answers among a set given a questions



LIVE
LEARNING | INTELLIGENCE | VISION | EMBODIEMENT

Meta

NEXT VISION
Spin-off of the University of Catania

Francesco Ragusa, Michele Mazzamuto, Rosario Forte, Irene D'Ambra, James Fort, Jakob Engel, Antonino Furnari, Giovanni Maria Farinella, Ego-EXTRA: video-language Egocentric Dataset for EXpert-TRAinee assistance, IEEE Winter Conference on Application of Computer Vision (WACV), 2026

WACV 2026
Tucson, Arizona • Mar 6 - 10, 2026

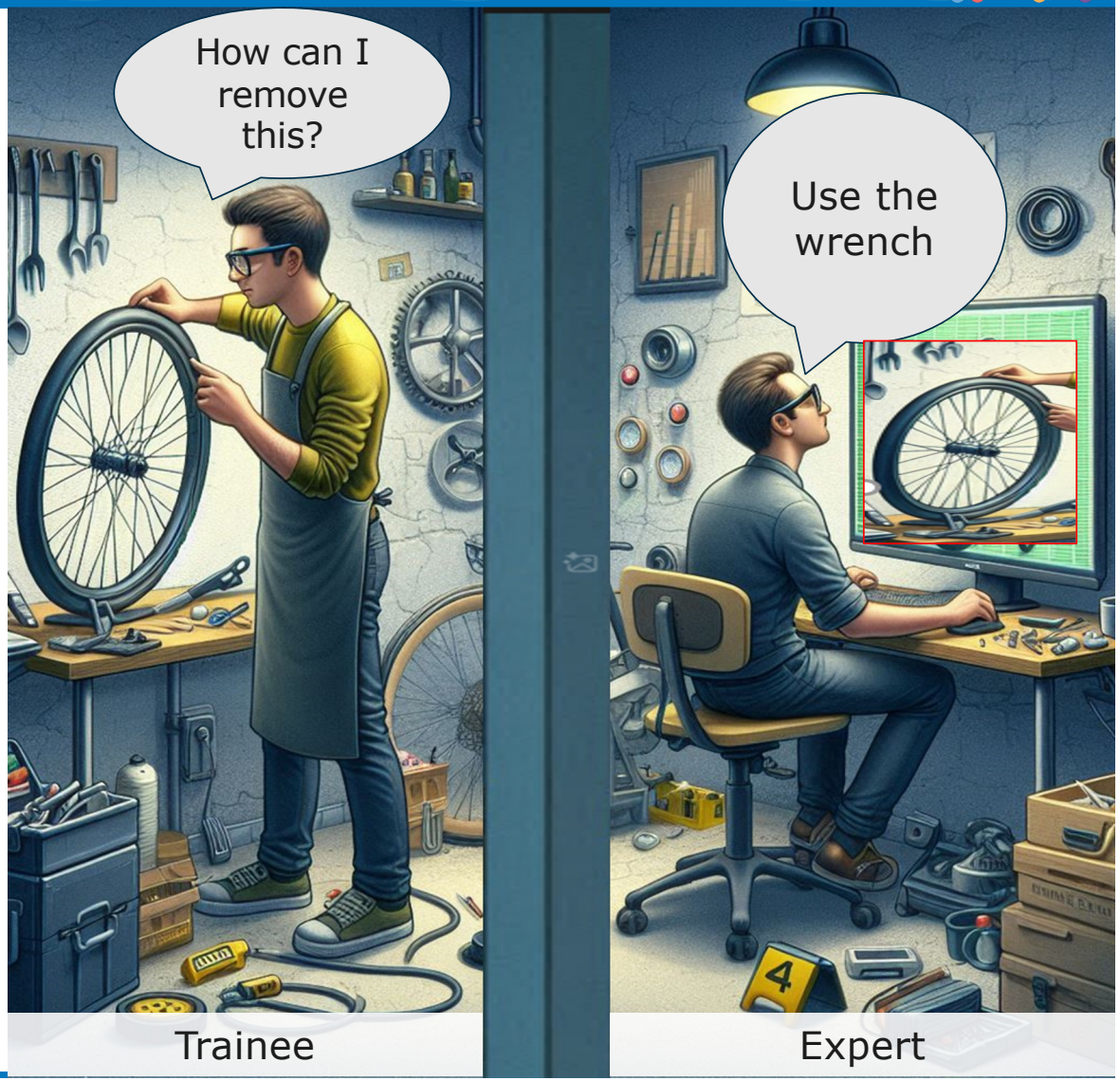
Ego-EXTRA: Dataset and Benchmark

Collection of egocentric videos of subjects engaging different procedural activities in which they are not expert or not very expert (i.e., Trainees);

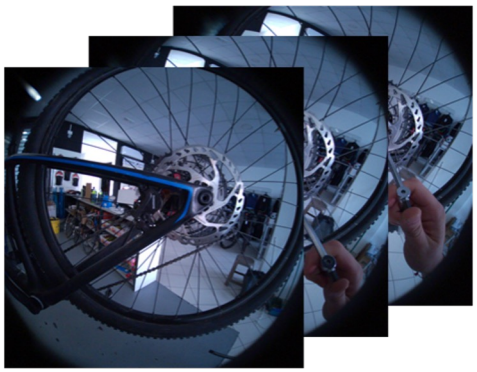
Collection of real conversations between trainees and experts which happen naturally during the procedure.

Benchmarking MLLM ability to select the correct answers among a set given a questions

Francesco Ragusa, Michele Mazzamuto, Rosario Forte, Irene D'Ambra, James Fort, Jakob Engel, Antonino Furnari, Giovanni Maria Farinella, Ego-EXTRA: video-language Egocentric Dataset for EXpert-TRAinee assistance, IEEE Winter Conference on Application of Computer Vision (WACV), 2026



Multiple-Choice Question Answering



Video Clip

"Do I need to worry that the wheel might fall?"

Trainee's question

Input

MiniGPT4-video → C

Video-LLama → A

Expert (GT) → A

A) "No, not at this moment. Now, hold it like that. "

B) "Maybe we should stop and secure everything again to be absolutely sure."

C) "No, but it's better to use additional supports or have someone assist you just in case."

D) "No, just let go and see if it stays in place."

F. Ragusa, M. Mazzamuto, R. Forte, I. D'Ambra, J. Fort, J. Engel, A. Furnari, G.M. Farinella: "Ego-EXTRA: Video-Language Egocentric Dataset for EXpert-TRAinee Assistance." WACV, 2026.

Results

	Model	Bike Workshop	Bakery	Assembly	Kitchen	Avg.
Language Only	Llama 3.1 Instruct 8B	20.20	25.71	20.11	25.00	21.30
	Llama 3.1 Instruct 70B	20.20	21.43	20.11	36.11	21.50
	Llama 3.3 Instruct Turbo	23.74	21.43	17.99	30.56	21.70
	Qwen 2.5 Instruct 72B	27.27	27.14	21.16	22.22	24.54
	DeepSeek-R1 Turbo	21.21	28.57	21.16	22.22	22.31
Video-Language	MiniGPT4-video	30.00	29.55	27.93	41.30	30.03
	LLaVa Video	27.78	38.57	29.10	27.78	29.82
	LLaVa-OneVision	**38.89**	**42.86**	**42.86**	**44.44**	**41.38**
	Qwen 2.5-VL	29.80	31.43	28.57	25.00	29.21
	Sample Human Baseline	87.50	90.91	100	81.82	89.65

Table 2. Results on the proposed VQA benchmark.

F. Ragusa, M. Mazzamuto, R. Forte, I. D'Ambra, J. Fort, J. Engel, A. Furnari, G.M. Farinella: "Ego-EXTRA: Video-Language Egocentric Dataset for EXpert-TRAinee Assistance." WACV, 2026.

What's Relevant in EgoVision? A top-down approach

An Outlook into the Future of Egocentric Vision

Chiara Plizzari* · Gabriele Goletto* · Antonino Furnari* · Siddhant Bansal* · Francesco Ragusa* · Giovanni Maria Farinella† · Dima Damen† · Tatiana Tommasi†

Politecnico di Torino · University of BRISTOL · Università di Catania

Abstract *What will the future be? We wonder!*
In this survey, we explore the gap between current research in egocentric vision and the ever-anticipated future, where wearable computing, with outward facing cameras and digital overlays, is expected to be integrated in our every day lives. To understand this gap, the article starts by envisaging the future through character-based stories, showcasing through examples the limitations of current technology. We then provide a mapping between this future and previously defined research tasks. For each task, we survey its seminal works, current state-of-the-art methodologies and available datasets, then reflect on shortcomings that limit its applicability to future research. Note that this survey focuses on software models for egocentric vision, independent of any specific hardware. The paper concludes with recommendations for areas of immediate explorations so as to unlock our path to the future always-on, personalised and life-enhancing egocentric vision.

Keywords Egocentric Vision, Future, Survey, Localisation, Scene Understanding, Recognition, Anticipation, Gaze Prediction, Social Understanding, Body Pose Estimation, Hand and Hand-Object Interaction, Person Identification, Summarisation, Dialogue, Privacy

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2.2 EGO-Worker 4
2.3 EGO-Tourist 5
2.4 EGO-Police 6
2.5 EGO-Designer 7
3 From Narratives to Research Tasks 8
4 Research Tasks and Capabilities 10
4.1 Localisation 10
4.2 3D Scene Understanding 14
4.3 Recognition 16
4.4 Anticipation 21
4.5 Gaze Understanding and Prediction 23
4.6 Social Behaviour Understanding 24
4.7 Full-body Pose Estimation 28
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*: Equal Contribution/First Author
†: Equal Senior Author
C. Plizzari, G. Goletto and T. Tommasi, Politecnico di Torino, Italy · A. Furnari, F. Ragusa and G. M. Farinella, University of Catania, Italy · S. Bansal and D. Damen, University of Bristol, UK. E-mail: Tatiana.Tommasi@polito.it

1 Introduction

Designing and building tools able to support human activities, improve quality of life, and enhance individuals' abilities to achieve their goals is the ever-lasting aspiration of our species. Among all inventions, digital computing has already had a revolutionary effect on human history. Of particular note is mobile technology, currently integrated in our lives through hand-held devices, i.e. *mobile smart phones*. These are nowadays the de facto for outdoor navigation, capturing static and moving footage of our everyday and connecting us to both familiar and novel connections and experiences.

However, humans have been dreaming about the next-version of such mobile technology — wearable computing, for a considerable amount of time. Imaginations

Immagine the Future

↓

Write Stories in Different Scenarios

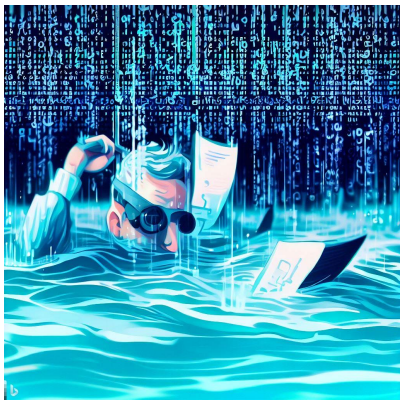
↓

Extract Important Tasks from the Stories

↓

Go in-depth with Tasks and Datasets

A lot of papers! (~ 30 years)



Rather than being extensive, we considered **seminal** and **state-of-the-art** works

Plizzari, C., Goletto, G., Furnari, A., Bansal, S., Ragusa, F., Farinella, G. M., Damen., D. & Tommasi, T. (2024). An Outlook into the Future of Egocentric Vision. International Journal of Computer Vision.

Università di Catania

An Outlook into the Future – Scenarios and Stories

EGO-HOME

Sam is finally home after a long day. EgoAI kept track of Sam's food intake and a tomato soup sounds like the best complementary nutrition

This way the tomato will cook evenly

A 3D projection of Remy helps with cooking

Audible 3D projection

Sam is impressed by how fun it is to cook with his 3D friend

Toaster reminder

EgoAI recommends some more spice

Waves hitting the shore look and sou...

While getting ready for bed, Sam feels an itch on the wrist that has annoyed him the whole day. EgoAI stores a picture of the injury and sends it to Sam's doctor for advice

EgoAI proposes a short clip from his day, but Sam decides not to share it

EGO-WORKER

EgoAI localises Marco and provides ruote instructions to reach his workstation for the day

EgoAI verifies if Marco is properly wearing the Personal Protection Equipment (PPE)

Where should I go today in the factory?

In the past, EgoAI guided Marco to the closest fire extinguisher during a fire

EgoAI passes a mes... from the m... to...

Since the me... devi...

...d turns off the IoT electrical socket while promptly alerting Marco

For the rest of the day, EgoAI validates Marco's work making sure all the procedures are properly and safely completed

By the end of the day, EgoAI checks Marco's feedback for improving future sessions

EGO-TOURIST

EgoAI prepares Claire a personalised and exciting one-day itinerary in Turin

EgoAI suggests an half-day visit to the Egyptian museum

EgoAI activates the 3D projection of Cleopatra to guide and interact with Claire

Claire feels transported to ancient Egypt

Claire asks Cleopatra for a good place for a pizza

...pizza place for ...ening Claire about the ...d various Italian monuments

...e next ... scheduled to arrive in 20 minutes

EgoAI offers a egocentric view from the chef who prepared her that delicacy

EgoAI suggests Claire a proper Italian coffee at a nearby cafè, sided by a slice of bunet, Turin-based dessert

EgoAI actively saved snapshots and videos of the day

EgoAI retrieves the closest souvenir shop based on Claire's taste and budget

EGO-POLICE

EgoAI is constantly pinpointing Judy's position and would send an alert to the headquarters if she encounters unusual events or dangerous situations

EgoAI helps Judy navigate through the shortest safe path to target places

Judy was able to swiftly arrest him

EgoAI has both thermal and multi-spectral sensors

Thanks to its sensors, EgoAI calculates a low risk for explosive content

Judy also appreciated the help of EgoAI when she had to manage an abandoned backpack

EgoAI projects a clear red circle around the backpack with the minimal stand-off distance

EgoAI connects Judy with the bomb squad and live-shares the observed scene

Thanks to EgoAI, all the relevant events are saved and transformed into a document with related images and video recordings

EgoAI guides Judy with exact instructions to grasp the backpack and open it

The sensitive information is properly identified and secured under admin rights to protect citizens' privacy

EGO-DESIGNER

EgoAI also suggests adding velvet couches on the right and a carved wooden table on the left

...access ...the database of the equipment warehouse; Stanley can search for the available pieces of furniture

EgoAI also allows Stanley to visualise how the actors should move in the space considering that there will be musicians in the middle of the room

EgoAI shares the scene with the actors. Through their own EgoAI, they are immersed inside the changing and moving 3D computer-generated environment

EgoAI assists make-up artists with advanced 3D modelling techniques to project guidelines on the actor's face while applying make-up

EgoAI also assists the director. He is able to preview the planned scene and light effects in real-time while shooting the scene

EGO-AI

From Narratives to Multimodal EgoAI Research Tasks

EGO-Home

1	3D Scene Understanding	1 2 3 4 7 8 9
2	Object and Action Recognition	1 5 6 10
4	Measuring Systems	6
5	Visual Question Answering	6
6 7	Summarisation and Retrieval	7
8 9 10	Full-Body\Hand Pose and Social Interaction	9
11	Medical Imaging	10
12	Messaging	10 11
6	Summarisation	11

EGO-Worker

13	Safety Compliance Assessment	1
14	Localisation and Navigation	2 5
12	Messaging	4
15	Hand-Object Interaction	5
16	Action Anticipation	6
17	Skill Assessment	7
5	Visual Question Answering	8
6	Summarisation	8

EGO-Tourist

18	Recommendation and Personalisation	1 2 8 9 10 11
1	3D Scene Understanding	2 3 4 5 6
19	Gaze Prediction	5
14	Localisation and Navigation	3 4 8 12
12	Messaging	7
5	Visual Question Answering	8
3 7	Action Recognition and Retrieval	11
6	Summarisation	13

EGO-Police

14	Localisation and Navigation	1 2
12	Messaging	1 3 11
3	Action Recognition	2 13
20	Person Re-ID	2 4
2 7	Object Detection and Retrieval	7
4	Measuring System	8 9
21	Decision Making	9
1	3D Scene Understanding	10
15	Hand-Object Interaction	12
6	Summarisation	13
22	Privacy	14

EGO-Designer

1	3D Scene Understanding	1 2 3 4 5 6 7 8
18	Recommendation	3
2 7	Object Recognition and Retrieval	3 4
8	Full-Body Pose Estimation	5 6
10	Social Interaction	6
19	Gaze Prediction	6
15	Hand-Object Interaction	7
12	Messaging	6 8

C. Plizzari, G. Goletto, A. Furnari, S. Bansal, F. Ragusa, G. M. Farinella, D. Damen, T. Tommasi, An Outlook into the Future of Egocentric Vision, *arXiv preprint arXiv:2308.07123*, 2023

Industrial Partners on Projects and and PhD Programs (since 2006)

Xenia SOFTWARE SOLUTIONS
2014 -

Meta facebook
2019 -

NVIDIA
2020 -

PHILIPS
2018 -

intel
2021 -

OSRAM
2016 -2018

TOYOTA
2024 -

life.augmented
2006 -

JOINTOPENLAB

TIM
2013 - 2018

linkverse
2019 - 2022

Centro Studi
Process Development & Applied Research
gruppo orizzonti holding
2015 - 2018

orangedev
2018 -

morpheos
2018 -

PARK SMART
2015 - 2019

NEXT VISION
2021 -

Selected Projects

2020-2023 (XENIA)

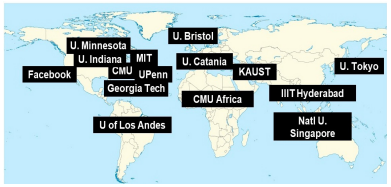
Localization, monitoring, and anticipation in industrial environment from First Person Vision

2020-2023 (XENIA)

Localization and attention estimation in cultural sites from First Person Vision

2019-2023 (Philips)

Personal Healthcare Interface Leveraging Human-Machine Natural Interactions from First Person Vision



2019-Today

(Meta - Facebook)

+3000 hrs of Egocentric Videos for Ego Tasks Benchmark

Other Projects supported by Industry

- Assistive Wearable Technologies and AR for Rehabilitation (AIAS 2021-2024)
- Long Term Egocentic Vision (INTEL – from Oct 2021)
- Procedural Understanding (Toyota – from 2024)

Projects can be found at https://iplab.dmi.unict.it/fpv/projects

Patent Accepted

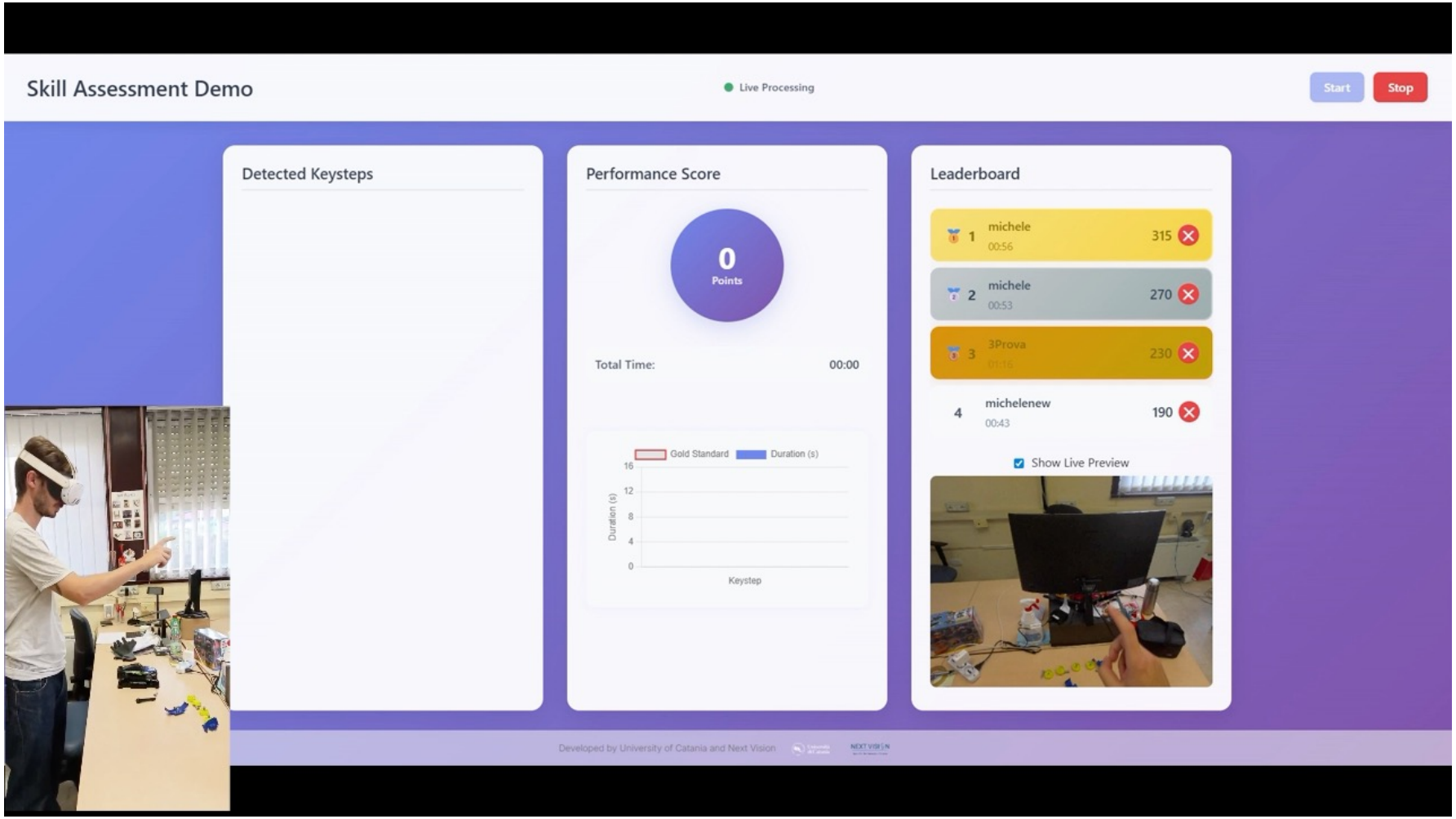
Number: 102020000027759

Acceptance date: 30/11/2022

Enhance Training

Improve Safety

Energy Saving


Skill Assessment Demo
Live Processing
Start
Stop
Detected Keysteps
Performance Score
0
Points
Total Time:
00:00
Gold Standard
Duration (s)
16
12
8
4
0
Duration (s)
Keystep
Leaderboard
1
michele
00:56
315
2
michele
00:53
270
3
3Prova
01:16
230
4
michelenew
00:43
190
Show Live Preview
Developed by University of Catania and Next Vision

Opening new direction
@ LIVE Research Group

Robot Learning by Seeing:
Towards Physical AI from Egocentric Human Experience
REC
Live Research Group with Toyota
EGOCENTRIC HUMAN EXPERIENCE
OBSERVE & RECORD
LEARN & UNDERSTAND
TRANSFER TO PHYSICAL ACTION
COLLABORATE INTELLIGENTLY

NEXT VISION

Spin-off of the University of Catania

https://www.nextvisionlab.it/

Artificial Intelligence to Assist Humans Where They Live and Work

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Founder & Chief Scientific Officer

Università di Catania

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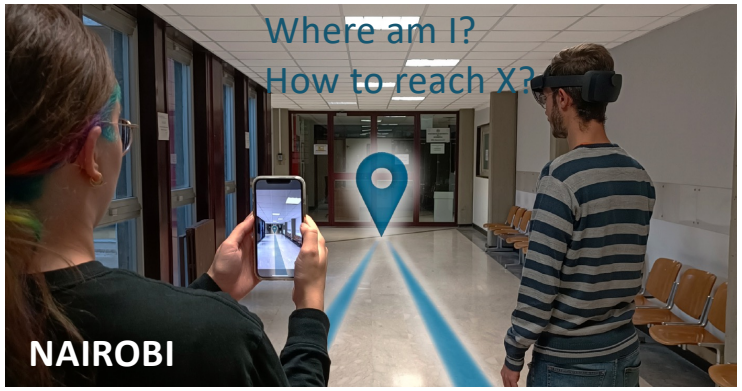
IMAGE PROCESSING LABORATORY

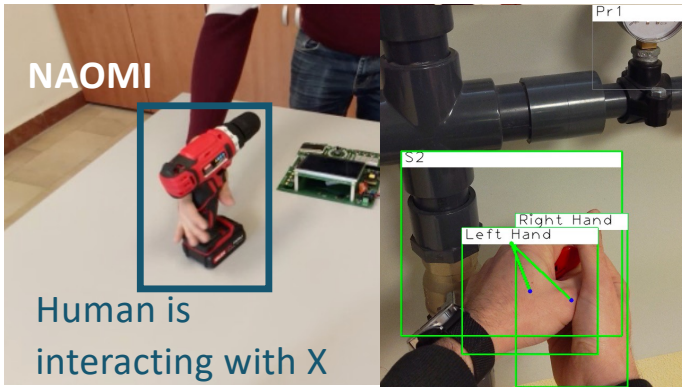
Some Solutions

1



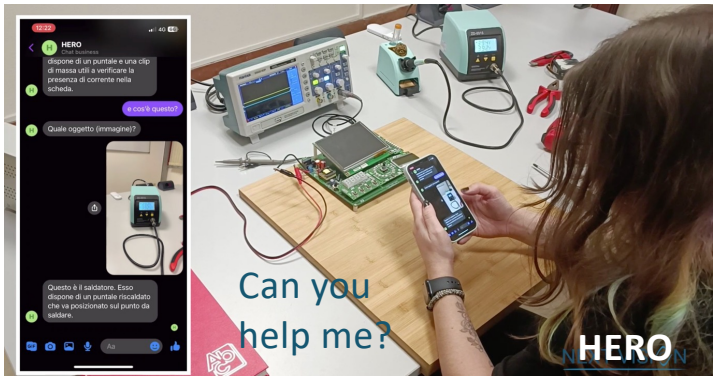
Localization of Humans and Navigation| Aware of Context

2



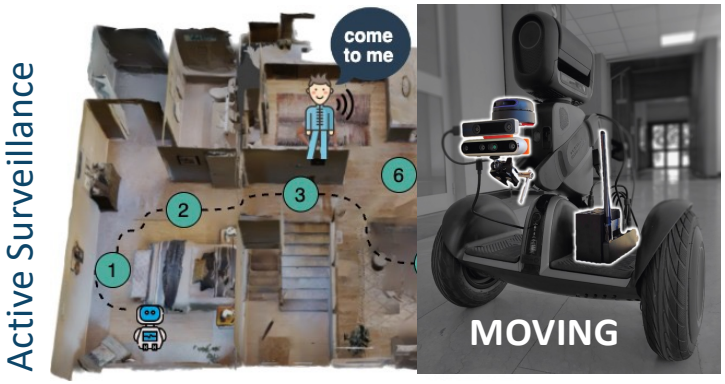
Human-Object Interaction | Aware of Objects

3



Conversational AI Agents | Aware of Human Needs and Intents

4



Autonomous AI Agents | Aware of Cooperation

NEXT VISION

Spin-off of the University of Catania

We are continuously searching for talents!

Several students have completed internships and theses in collaboration with Next Vision.

One of the industrial seminars will be organized in collaboration with Next Vision.